Perfect play with spatial quantum strategies is Σ_2^0-complete

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Abstract

When does a finite nonlocal game admit perfect spatial quantum play? We prove that this decision problem is complete for the second existential level of the arithmetical hierarchy. The same classification holds when a perfect strategy must have finite marginal entropy. Hardness persists among games whose finite-dimensional value is one and which have no finite-dimensional perfect strategy. A finite clock-phase order supplies the existential witness: normality forces a finite order on a nonzero sector, while the strategy chooses the order.

For the upper bound, we normalize the largest Schmidt coefficient and use compact operator sandwiches. Fixing a lower bound on the largest Schmidt probability gives a co-recursively enumerable stratum. No computable bound selects a sufficient stratum for every game. Exact attainment of a rational Bell bound, of an unspecified supremum, and of a specified scalar value is also Σ_2^0-complete. The hardness proof extends the modular-shift and rational-row toolkit of our earlier entropy-game paper to four calibrated bridges and a deterministic computation on rooted finite squares.

AI systems assisted with this work. The authors take responsibility for its claims; see "How this work was done." This preprint has not been peer reviewed.

Changes in version 1.1.0. No theorem statement has changed. Section 4 adds a one-page overview, "From a presentation to a game," and writes out the infinite-dimensional steps in the proofs of Lemmas 4.1–4.4 in full. Section 10.3 replaces the questions about a von Neumann type ladder by the results of [6, 5, 7]. Section 10.4 adds that perfect spatial play attainable only with infinite entropy is $D_2(\Sigma_2^0)$-complete [7]. Section 7 and Appendix A state that the only compiled game is the minimal compiler instance and that the full reduction game was not compiled.

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1 Introduction

When does a finite game admit a strategy that wins every round? Approaching certainty and attaining certainty are different questions. They also depend on the model of quantum strategy. Table 1 puts the distinction first. The spatial entry is the result of this paper.

Decision problem	Complete class	Source
Perfect finite-dimensional strategy	Σ_1^0	[11]
Perfect commuting strategy	Π_1^0	[21, 17]
Finite-dimensional value equals one	Π_2^0	[17]
Perfect spatial strategy	Σ_2^0	This paper

Table 1: Finite rational Boolean games; computable many-one reductions. A value is a supremum. A perfect strategy attains one.

The finite-dimensional and spatial models use tensor products. The latter allows an arbitrary normal state on infinite-dimensional local Hilbert spaces. The commuting model instead allows the two measurement algebras to act on one Hilbert space and commute. Its compact state-space description makes perfect commuting play equivalent to commuting value one. Spatial play has no such compactness in normalized states.

Our upper bound keeps one Schmidt coefficient from disappearing. After that coefficient is scaled to one, the amplitude operator belongs to a norm-compact family of compact operators. Losing probabilities become homogeneous zero equations. These survive a limit even when some squared Schmidt mass escapes. The surviving state can be normalized again.

The hardness mechanism goes in the other direction. A finite Weyl clock commutes with a trace-class density. Each positive density eigenspace is finite-dimensional, so a determinant forces the clock phase to have finite order. The strategy chooses that order. A deterministic computation reads it and checks an infinite sequence of decidable conditions. This is the existential quantifier, and no computable bound on the game can capture its size. The modular-shift toolkit comes from our earlier paper [8]; here four calibrated bridges connect it to the computation.

Theorem A (spatial perfection). *Deciding whether a finite rational* $0/1$ *game has a perfect normal spatial strategy is* Σ_2^0*-complete. The same holds when a perfect strategy with finite marginal entropy is required. Hardness holds under the promise* $\omega_q(G) = 1$*, for a family in which no instance has a perfect finite-dimensional strategy.*

Finite marginal entropy can be required on both sides. For pure strategies the two marginal entropies agree. For a perfect mixed strategy with one finite marginal entropy, a pure component has finite entropy and is still perfect; see Section 3.

Theorem B (Bell attainment). *For rational Bell functionals, each of the following is* Σ_2^0*-complete in the normal spatial model: attainment of a supplied rational global upper bound; attainment of the supremum, with no bound supplied; and attainment of a supplied rational scalar value. In the first problem, validity of the supplied bound can either be promised or required as part of the decision problem.*

Theorem C (compactness and Schmidt mass). *Uniformly for integers $\kappa \geq 1$, the existence of a perfect pure spatial strategy with largest Schmidt probability at least $1/\kappa$ is a Π^0_1 property. Their union is $\mathrm{PERF}_{\mathrm{qs}} \in \Sigma^0_2$. No total computable function $\kappa(G)$ bounds a sufficient stratum for every spatially perfect game. If $\omega_q(G) = 1$ and no perfect spatial strategy exists, the largest Schmidt probability tends to zero along every sequence of pure spatial strategies whose winning probabilities tend to one.*

1.1 Prior work and contribution

The finite-dimensional entry in Table 1 uses perfect completeness in Theorem 12.6 of Ji, Natarajan, Vidick, Wright and Yuen [11], not just a gap statement about a supremum. Upper membership follows by a dimension search and real quantifier elimination. Slofstra's embedding theorem [21] and the account of Mousavi, Nezhadi and Yuen [17] give the commuting entry. Their value-one theorem gives the third entry. The commuting upper bound uses the convergent moment hierarchy of Navascués, Pironio and Acín [18]. The operator-solution framework for binary constraint system games originates with Cleve and Mittal [3]; Cleve, Liu and Slofstra gave the commuting-operator extension [2]. Slofstra also established nonclosure of finite-dimensional quantum correlations [20]. These are distinct earlier results. Lin's $\mathsf{MIP}^{co} = \mathsf{coRE}$ theorem concerns gap approximation in the commuting model [13]; it does not classify spatial attainment.

Infinite-dimensional spatial correlations were constructed by Coladangelo and Stark [4]. Kim, Paulsen and Schafhauser prove that perfect spatial play collapses to perfect finite-dimensional play for synchronous games [12, Corollary 3.11]. Lupini and coauthors establish the corresponding equivalence for imitation games [14, Corollary 5.4]. Our games retain nonsynchronous bridge questions. No YES instance is synchronous or an imitation game, since either condition would imply finite-dimensional perfect play.

Mančinska's flagged magic-square construction supplies the bridge geometry [15]. The normal bridge, shared-label refinement, graph products and Boolean rational-row tests were developed in [8, Lemmas 4.1 and 4.2, Theorem B, Eq. (32)]. The versions needed here are restated and proved in Section 4. That section extends the assembly from a particular presentation to four bridges and arbitrary finite lists of endpoint labels. The rooted four-ary computation and the finite-period reader then give the hardness reduction. The compact-sandwich upper bound does not use this compiler.

Fu, Miller and Slofstra study fixed-scenario correlation membership [10]. A remark there asserts, without proof, that fixed-scenario membership in C_q and C_{qs} is in RE; Section 10 records a conditional tension with Theorem A. We do not identify prescribed-table membership with game perfection.

To our knowledge, Theorems A and B give the first Σ^0_2-completeness classifications of these spatial attainment problems. This priority statement is provisional. It rests on the cited hierarchy and correlation papers, their nearby results, and a targeted literature search. It is not an exhaustive bibliographic certificate.

$$\begin{gathered}
\text{Presentation} \xrightarrow{\text{Lemma 4.4}} \text{finite game } G_e \\
\text{Perfect normal strategy} \longrightarrow \text{normal realization} \\
\longrightarrow \text{Fock tree} \longrightarrow \text{finite clock sector} \\
\longrightarrow \text{deterministic tableau} \longrightarrow \forall t\ R(e,n,t) \\
\text{Witness } n \longrightarrow \text{Fock/Weyl model} \\
\longrightarrow \text{scaffold, frame, row ancilla} \longrightarrow \text{perfect strategy}
\end{gathered}$$

Figure 1: Dependencies of the hardness proof. Compactness and Bell attainment have separate upper-bound arguments.

1.2 Proof route

Figure 1 summarizes the hardness proof. Section 3 proves Theorem C and the finite-entropy upper bound. Section 4 gives the finite game compiler. Sections 5–7 encode a chosen finite witness and all its checks. Section 8 proves the value-one promise with four explicit boundary defects. Section 9 proves Theorem B. The verification appendix records what the finite programs check.

2 Preliminaries

A finite rational Boolean game is a finite list of questions and answers, a rational distribution $\pi(x, y)$, and a predicate $V(a, b \mid x, y) \in \{0, 1\}$. A spatial strategy consists of positive operator-valued measurements (POVMs) $E_{a|x}$ and $F_{b|y}$ and a normal state, represented by a positive trace-class density ρ of trace one, on $\mathcal{H}_A \otimes \mathcal{H}_B$. Its winning probability is

$$\sum_{xyab} \pi(x, y)V(a, b \mid x, y) \operatorname{Tr} \rho(E_{a|x} \otimes F_{b|y}).$$

We write C_q, C_{qs}, C_{qc} for finite-dimensional, normal spatial, and commuting correlations. For a strategy $\mathcal{S}$, write $\omega(G, \mathcal{S})$ for the displayed winning probability and $\mathrm{loss}_G(\mathcal{S}) = 1 - \omega(G, \mathcal{S})$. The set $\mathrm{PERF}_{\mathrm{qs}}$ consists of games with a perfect normal spatial strategy. Its stratum $\mathrm{PERF}_{\mathrm{qs}}(\kappa)$ requires a perfect pure strategy whose largest Schmidt probability is at least $1/\kappa$. The set $\mathrm{PERF}_{\mathrm{finS}}$ requires a perfect strategy with finite marginal entropy. The value ω_q is the supremum over C_q. Finite Schmidt truncations show that C_q and C_{qs} have the same closure and the same supremum for every Bell functional. Entropy uses logarithms to base two: $S(h) = -\operatorname{Tr}(h \log_2 h)$ and $H(p) = -\sum_i p_i \log_2 p_i$, with $0 \log_2 0 = 0$. A projection-valued measurement (PVM) is a POVM consisting of orthogonal projections. Write $|\Phi_d\rangle = d^{-1/2} \sum_{j=0}^{d-1} |j, j\rangle$ for a finite maximally entangled vector. Vectorization of an operator T is $|T\rangle\rangle = \sum_{ij} T_{ij}|i, j\rangle$ when this sum is square summable. A purification is a unit vector with the given state as its partial trace over an added environment.

The set Σ_2^0 consists of predicates $\exists n \, \forall t \, R(e, n, t)$ with decidable R. The class Π_1^0 consists of universal decidable predicates. All encodings and reductions in this paper are effective finite encodings and computable many-one reductions. Completeness is for an unrestricted finite scenario, not one fixed question-and-answer size.

Fact 2.1 (pure components). If a mixed normal strategy is perfect, write $\rho = \sum_j \lambda_j |\psi_j\rangle\langle\psi_j|$. Each losing event is positive. Its expectation vanishes in every component with $\lambda_j > 0$, so every such component is perfect with the same measurements. Trace-class marginals have separable support. Thus all arguments may be made on separable state supports.

Fact 2.2 (matching and support). For effects on opposite sides, perfect binary matching gives, on a purification Ω,

$$0 = \langle E + F - 2EF\rangle = \langle (E - F)^2 + (E - E^2) + (F - F^2)\rangle.$$

Each positive term annihilates Ω. If T is the coefficient operator of the pure vector on the chosen Schmidt cut, matching gives $ET = TF^{\mathsf{T}}$; dense coefficient range first makes the local state support reducing. On that support the effects are projections and commute with the marginal density. A positive operator B with $\operatorname{Tr}(hB) = 0$ for faithful h is zero, since $B^{1/2}h^{1/2} = 0$ and $h^{1/2}$ has dense range.

Fact 2.3 (sharp compressions and joint measurements). If PEP is a projection for an effect E and support projection P, positivity shows that E acts as identity on its range and zero on its kernel inside $P\mathcal{H}$. Hence E reduces the support. A POVM with sharp coarse marginals is their joint spectral measurement: each fine effect is supported in the corresponding marginal ranges, and summation gives the commuting products. These facts apply after compression before projectivity is used.

Fact 2.4 (synchronization and forbidden events). We call a projection synchronized when its binary question is perfectly matched to the corresponding opposite question. If synchronized projections satisfy $P_A^{\mathrm{sync}}\Omega = P_B^{\mathrm{sync}}\Omega$ and a forbidden event gives $Q_A P_B^{\mathrm{sync}}\Omega = 0$, dense Schmidt range gives the local identity $Q_A P_A^{\mathrm{sync}} = 0$. The arguments work on either party and on purifications of mixed states.

Table 2 collects the notation.

Table 2: Notation used throughout the proof. Distinct fonts and subscripts separate question names, state spaces and operators.

Symbol	Meaning
κ, D_{Sch}	Schmidt stratum integer and normalized amplitude operator
$D_{\mathrm{esc},n}$	escaping-mass example in Section 10
$\mathsf{R}_i, \mathsf{C}_j, \mathsf{T}_x, \mathsf{B}_y$	flagged magic-square questions
$P_\circ, \mathcal{K}$	root projection and root multiplicity space
$P_{\mathrm{mag}}, h_{\mathrm{mag}}$	magic support and its density block
$T_f, T_{\mathrm{mag}}, T_{\mathrm{cal}}, \Delta$	flag/magic coefficient blocks, their calibrated diagonal and its sign reflection
$U_x, U_y; P_x, P_y$	cyclic coordinate shifts; zero-column and zero-row projections
$J_{\mathrm{W}}, K_{\mathrm{W}}, Z_{\mathrm{clk}}$	Weyl pair and central clock phase
$W, C_{\mathrm{ctr}}, P_{\mathrm{ret}}, T_{\mathrm{inv}}$	coordinate clock, counter, return pulse and bounded reader inverse
P_c	tableau symbol projections, $c \in \Gamma$
e_g, M_g, w_g, v	scaffold corners, magic blocks, single bridges and their chain
L_r	decoded copy of a typed label L at corner e_r
$\mathcal{H}_{\mathrm{base}}, F_i, G_{ij}$	graph base space, frame projections and graph projections
B_{ij}, s_{par}	parameter-binding projections and its reflection label
$P_\triangle, \Pi_t, \mathcal{I}_t$	triangle projection, row-column partition and column isometries
$h_{\mathrm{old}}, h^{[m]}$	marginal before row ancilla; truncated Fock/Weyl density
C_{abs}	sum of absolute Bell coefficients
$\mathcal{S}, \omega(G, \mathcal{S})$	strategy and its winning probability

3 The upper bound

3.1 Normalize one Schmidt coefficient

For a pure strategy write $\psi = \sum_i s_i |i,i\rangle$ with $s_1 \geq s_2 \geq \cdots$ and $\sum_i s_i^2 = 1$. Set $d_i = s_i/s_1$ and $D_{\mathrm{Sch}} = \mathrm{diag}(d_i)$. For some integer κ,

$$d_1 = 1, \qquad d_i \downarrow 0, \qquad \operatorname{Tr} D_{\mathrm{Sch}}^2 \leq \kappa. \tag{1}$$

For positive effects E, F, vectorization gives

$$p(E,F) = \frac{\|\sqrt{E}\, D_{\mathrm{Sch}} \sqrt{F^{\mathsf{T}}}\|_{\mathrm{HS}}^2}{\operatorname{Tr} D_{\mathrm{Sch}}^2}, \qquad p(E,F) = 0 \iff E D_{\mathrm{Sch}} F^{\mathsf{T}} = 0.$$

To prove the reverse implication, $E D_{\mathrm{Sch}} F^{\mathsf{T}} = 0$ sends $D_{\mathrm{Sch}} \operatorname{ran} F^{\mathsf{T}}$ into $\ker E$. The closures of the ranges of F^{T} and $\sqrt{F^{\mathsf{T}}}$ agree, and $\ker E = \ker \sqrt{E}$. Continuity gives the square-root sandwich identity. The forward implication follows by multiplying the zero square-root sandwich by the square roots.

Let P_n be the first n coordinate projection. Monotonicity in (1) gives

$$\|D_{\mathrm{Sch}} - P_n D_{\mathrm{Sch}} P_n\| = d_{n+1} \leq \sqrt{\kappa/(n+1)}. \tag{2}$$

Consequently the family in (1) is compact in operator norm: take coordinatewise limits and use this common tail estimate; Fatou preserves the squared-sum bound. Finite families of POVMs on ℓ^2 are compact in the weak operator topology. Positivity and the finite sum-to-identity relations survive these limits.

Lemma 3.1 (Compact sandwiches)**.** *If $D_{\mathrm{Sch},j} \to D_{\mathrm{Sch}}$ in norm with compact D_{Sch}, and uniformly bounded E_j, F_j converge weakly to E, F, then $E_j D_{\mathrm{Sch},j} F_j$ converges weakly to $E D_{\mathrm{Sch}} F$.*

Proof. For fixed v, compactness makes $D_{\mathrm{Sch}} F_j v \to D_{\mathrm{Sch}} F v$ in norm. Pair this with the weakly convergent vectors $E_j^* u$. The replacement of D_{Sch} by $D_{\mathrm{Sch},j}$ has norm error tending to zero. □

3.2 An effective finite predicate

Let $\mathcal{L}_G$ be the losing tuples with $\pi(x,y) > 0$. Define $R(G,\kappa,n)$ by the existence of n-dimensional POVMs and real numbers $1 = d_1 \geq \cdots \geq d_n \geq 0$, $\sum_i d_i^2 \leq \kappa$, such that

$$|(E_{a|x} D_{\mathrm{Sch},n} F_{b|y}^{\mathsf{T}})_{ij}|^2 \leq \kappa/(n+1) \quad ((x,y,a,b) \in \mathcal{L}_G;\ i,j \leq n). \tag{3}$$

This is semialgebraic over the rationals. Splitting complex entries into real and imaginary parts and using principal minors for positivity makes real quantifier elimination a decision procedure.

A perfect strategy with $s_1^2 \geq 1/\kappa$ supplies all these witnesses. Compress its POVMs. Since P_n commutes with D_{Sch} and $E D_{\mathrm{Sch}} F^{\mathsf{T}} = 0$, the finite sandwich is $-P_n E (D_{\mathrm{Sch}} - D_{\mathrm{Sch},n}) F^{\mathsf{T}} P_n$, with norm bounded by (2).

Conversely choose a witness for every n. Extend $D_{\mathrm{Sch},n}$ by zero and each POVM by a fixed deterministic POVM on the complement. Pass to a subsequence with $D_{\mathrm{Sch},n} \to D_{\mathrm{Sch}}$ in norm and all effects weakly convergent. For every fixed matrix entry, (3) and Lemma 3.1 give $E D_{\mathrm{Sch}} F^{\mathsf{T}} = 0$. We retain

$$1 \leq \operatorname{Tr} D_{\mathrm{Sch}}^2 \leq \kappa, \qquad d_1 = 1.$$

The normalized vectorization of D_{Sch} is a perfect normal strategy with largest Schmidt probability $1/\operatorname{Tr} D_{\mathrm{Sch}}^2 \geq 1/\kappa$. Therefore

$$G \in \mathrm{PERF}_{\mathrm{qs}}(\kappa) \iff \forall n\ R(G,\kappa,n), \qquad \mathrm{PERF}_{\mathrm{qs}} = \bigcup_{\kappa\geq 1} \mathrm{PERF}_{\mathrm{qs}}(\kappa). \tag{4}$$

This proves the upper membership in Theorems A and C.

3.3 No effective choice of stratum; vanishing largest weight

If a total computable $\kappa(G)$ were sufficient for every YES game, (4) would put spatial perfection in Π_1^0. Perfect completeness of [11, Theorem 12.6] already makes it Σ_1^0-hard. NO instances of that map have spatial value at most $1/2$, because the finite-dimensional and spatial suprema agree. Thus the same map reduces the halting set to spatial perfection. The halting set cannot be Π_1^0, so such a function does not exist.

Now suppose pure spatial strategies approach perfect play while $p_{\max} \geq 1/\kappa$ on a subsequence. Normalize their largest Schmidt amplitudes. For a losing event,

$$\|ED_{\mathrm{Sch}}F^{\mathsf{T}}\|_{\mathrm{HS}} \leq \|\sqrt{E}D_{\mathrm{Sch}}\sqrt{F^{\mathsf{T}}}\|_{\mathrm{HS}} = \sqrt{\operatorname{Tr} D_{\mathrm{Sch}}^2\, p(E,F)}.$$

Positive question weights force every such probability to zero. The same compact limit gives a perfect spatial strategy. Contraposition proves the final assertion of Theorem C, including sequences of finite-dimensional pure strategies. Equivalently their Schmidt min-entropy tends to infinity.

3.4 Finite entropy and rational slack

For sorted Schmidt probabilities p_i, $p_i \leq 1/i$. If $H(p) \leq B$, then

$$\delta_n = \sum_{i>n} p_i \leq B/\log_2(n+1). \tag{5}$$

The normalized first-n truncation majorizes p, so has entropy at most B. Its pure-state trace distance from the original state is $\sqrt{\delta_n}$. This bounds the change in any winning probability. Sorted distributions with entropy at most B are compact in ℓ^1: (5) gives tightness and entropy is lower semicontinuous. Their square roots are compact in ℓ^2. Coupled with weak compactness of POVMs, this gives compactness of the resulting correlations.

For effectivity use slack rather than treating entropy as a polynomial. Fix integers $B, n \geq 1$. Choose computably a rational $\eta_{B,n} > 0$ converging to zero with $\eta_{B,n} > \sqrt{B/\log_2(n+1)}$. For every $k \geq 1$ compute a rational polynomial $H_{n,k}$ such that

$$\sup_{p\in\Delta_n} |H_{n,k}(p) - H(p)| \leq 1/k, \qquad \Delta_n = \{p_i \geq 0 : \sum_i p_i = 1\}.$$

Such polynomials are effective, for example by Bernstein approximation with rationally rounded coefficients to the computable continuous function $-t\log_2 t$ on $[0,1]$, with error $1/(nk)$ per summand. At precision k require sorted $p \in \Delta_n$, real amplitudes $s_i \geq 0$ with $s_i^2 = p_i$, finite POVMs, and

$$\mathrm{loss}_G(s,E,F) \leq \eta_{B,n}, \qquad H_{n,k}(p) \leq B + 1/n + 1/k.$$

The loss is a rational polynomial in the amplitudes and real and imaginary parts of the effects. Real quantifier elimination therefore decides this finite relaxation. Exact entropy feasibility implies every relaxation. Conversely, if every k-relaxation is feasible, compactness

of this finite variable domain yields a subsequential limit; the uniform approximation error gives $H(p) \le B + 1/n$ and the closed loss constraint survives. Thus exact stage feasibility is equivalent to feasibility of all these rational relaxations, even though their sets need not be nested. Flattening $\forall n \forall k$ gives a uniform Π^0_1 predicate for the stratum $\mathrm{PERF}_{\mathrm{finS}}(B)$ with supplied entropy budget B, defined by a perfect pure strategy with $H(p) \le B$.

A strategy of finite entropy satisfies the stages for an integer B above its entropy. Conversely stage witnesses have a common entropy bound $B+1$, hence the tail compactness just proved. A limit is perfect and has entropy at most B. Thus $\mathrm{PERF}_{\mathrm{finS}} \in \Sigma^0_2$. For mixed states, concavity of marginal entropy gives $\sum_j \lambda_j S((\psi_j)_A) \le S(\rho_A)$. If the right side is finite, a positive-weight component has finite entropy and is perfect. This proves the claimed equivalence of the entropy conventions.

4 A finite compiler with four bridges

The modular operators below are not synchronized. A synchronized operator commutes with the marginal density, so it cannot itself change a density eigenvalue. We retain conditional flagged magic-square questions to supply that change. The density centralizer is the set of bounded operators commuting with the density. Everything submitted to the graph compiler is in it.

We first restate the bridge, label and rational-row tools from [8, Lemmas 4.1 and 4.2, Theorem B, Eq. (32)], with their proofs. These arguments concern arbitrary normal supports and finite word lists. We then give the four-bridge assembly. No product of two modular transports is submitted as a centralizer word.

From a presentation to a game

This overview says what the compiler does. The proofs follow in Sections 4.1–4.5.

Input. An admissible presentation (Section 4.4) is a finite rational list: four contraction symbols A_α, a root projection $P_\circ$, finitely many typed centralizer labels and finitely many rational $*$-polynomial equations. Its intended solutions are normal realizations: bounded operators on a Hilbert space $\mathcal{H}_0$ with a faithful trace-class density h_0, such that the labels commute with h_0, the equations hold, and $h_0 A_\alpha = A_\alpha h_0/16$.

Output. A finite game with a rational question distribution and a 0/1 predicate. Its perfect normal spatial strategies and the normal realizations of the presentation correspond in both directions (Lemma 4.4).

The pipeline.

1. *Labels become unitaries.* A projection P is encoded by $2P - I$, a positive contraction T_+ by a unitary u_+ with $T_+ = (2I + u_+ + u_+^*)/4$, and a general contraction by an average of two unitaries.
2. *A scaffold carries the factor* 16. Seventeen synchronized corners and four conditional flagged magic-square bridges certify partial isometries w_g with $hw_g = 2w_g h$ (Lemma 4.1). Their product v maps the source corner e_0 onto e_4 with $hv = 16vh$, and the cover fixes the source mass 1/121.
3. *Labels travel through the bridges.* Corner copies, block unitaries and shared-question refinements (Lemma 4.2) force $L_4 v = vL_0$ for every transported label L.
4. *The modular operators disappear.* Each creation symbol is written $A_\alpha = v^* a_\alpha$ with a centralizer contraction a_α from e_0 to e_4. Every application equation becomes an equation among centralizer labels in which v does not occur.
5. *Words become projections.* A qutrit frame, graph projections and triangle projections certify actual products of unitary labels. A rational equation among words becomes a

rational linear equation among synchronized projections (Section 4.2).

6. *Linear equations become Boolean tests.* Each such equation is enforced by finitely many orthogonality tests against a certified scalar matrix algebra (Lemma 4.3).
7. *Assembly.* All questions are kept, the question sets are padded, and question pairs are sampled uniformly; unspecified pairs are accepted.

Soundness. In a perfect normal strategy every losing event of positive weight has probability zero. This makes each synchronized event a projection on the state support that commutes with the marginal density, and it turns each submitted equation into an operator identity on that support. The bridges then give the realization on the source corner, with density $h_0 = 121e_0he_0$. No step inverts a density.

Completeness. A realization is copied into the scaffold density (20). Contractions are written as averages of unitaries inside the finite-dimensional positive eigenspaces of that density. A qutrit frame and a finite maximally entangled row ancilla are appended, and Bob uses transposes in an eigenbasis of the density. Every test then holds exactly. Only finite-dimensional factors are added, so finite entropy is preserved.

What was compiled. Section 4.5 bounds the loss when the application equations hold only approximately, and Section 7 bounds the size of the reduction game. The one compiled instance, described in Appendix A, is minimal: four bridges, one modular isometry and no transported labels. The full reduction game was not compiled.

Normal states, supports and exact cancellation

The proofs in this section use only the following facts. They make Facts 2.1–2.4 explicit for arbitrary, possibly nonseparable, Hilbert spaces. None of them uses a dimension count, a determinant or a bounded inverse of a density.

(N1) Normal states and purifications. A normal state on $\mathcal{H}_A \otimes \mathcal{H}_B$ is a positive trace-class operator ρ of trace one. By the spectral theorem for compact operators, $\rho = \sum_j \lambda_j |\psi_j\rangle\langle\psi_j|$ with countably many $\lambda_j > 0$. Let J be the countable index set, put $\mathcal{H}_{\mathrm{env}} = \ell^2(J)$, and put $\Omega = \sum_j \lambda_j^{1/2} \psi_j \otimes |j\rangle$. Then

$$\operatorname{Tr} \rho(E \otimes F) = \langle \Omega, (E \otimes F \otimes I_{\mathrm{env}})\Omega \rangle$$

for all bounded E, F, and no measurement acts on $\mathcal{H}_{\mathrm{env}}$. The marginals $h_A = \operatorname{Tr}_B \rho$ and $h_B = \operatorname{Tr}_A \rho$ are positive trace-class operators of trace one. Write P_A, P_B for their support projections, the projections onto the closures of their ranges; these closures are separable. The restriction of h_A to $P_A\mathcal{H}_A$ is injective with dense range. Its positive eigenspaces are finite-dimensional and span a dense subspace of $P_A\mathcal{H}_A$. The same holds for h_B. If P_A has infinite rank, the inverse of h_A on its range is unbounded; it is never used.

(N2) Coefficient operators. Put $\mathcal{H}_{B,\mathrm{env}} = \mathcal{H}_B \otimes \mathcal{H}_{\mathrm{env}}$. Fix orthonormal bases (b_j) of $\mathcal{H}_B$ and (e_ℓ) of $\mathcal{H}_{\mathrm{env}}$, and use their tensor-product basis $(f_{j\ell}) = (b_j \otimes e_\ell)$ of $\mathcal{H}_{B,\mathrm{env}}$, and let $Y^{\mathsf{T}} = CY^*C$ be the transpose of $Y \in B(\mathcal{H}_{B,\mathrm{env}})$, where C is the complex conjugation fixing every $f_{j\ell}$. Bob's transpose F^{T} is defined in the basis (b_j). Transposition preserves norms, adjoints, positivity and projections, reverses products, and satisfies $(F \otimes I_{\mathrm{env}})^{\mathsf{T}} = F^{\mathsf{T}} \otimes I_{\mathrm{env}}$. The map $T \mapsto \sum_{j,\ell} T f_{j\ell} \otimes f_{j\ell}$ is a unitary from the Hilbert–Schmidt operators $\mathcal{H}_{B,\mathrm{env}} \to \mathcal{H}_A$ onto $\mathcal{H}_A \otimes \mathcal{H}_{B,\mathrm{env}}$. If it sends T_Ω to Ω, it sends $XT_\Omega Y^{\mathsf{T}}$ to $(X \otimes Y)\Omega$, and $T_\Omega T_\Omega^* = h_A$. We call T_Ω the coefficient operator. Identities such as $ET = TF^{\mathsf{T}}$ below are coefficient forms of vector identities, and we state both forms where it helps. For a mixed state this is the cut $A \mid B\,\mathrm{env}$, and every Bob operator enters as $F \otimes I_{\mathrm{env}}$.

(N3) Exact cancellation. For $X \in B(\mathcal{H}_A)$,

$$\|(X \otimes I)\Omega\|^2 = \mathrm{Tr}(h_A X^* X) = \|X h_A^{1/2}\|_{\mathrm{HS}}^2.$$

Hence $(X \otimes I)\Omega = 0$, equivalently $XT_\Omega = 0$, holds exactly when $XP_A = 0$. Indeed the range of $h_A^{1/2}$ is dense in $P_A\mathcal{H}_A$, and a bounded operator that vanishes on a dense subspace vanishes on its closure. Likewise $(I \otimes F \otimes I_{\mathrm{env}})\Omega = 0$, equivalently $T_\Omega(F \otimes I_{\mathrm{env}})^{\mathsf{T}} = 0$, holds exactly when $FP_B = 0$, because its squared norm is $\mathrm{Tr}(h_B F^* F)$. These are the dense-range cancellations used throughout. The same statements hold for any vector in place of Ω, with the supports of its own marginals.

(N4) Mirrors. Call $X \in B(\mathcal{H}_A)$ and $Y \in B(\mathcal{H}_B)$ *mirrors* if $(X \otimes I)\Omega = (I \otimes Y \otimes I_{\mathrm{env}})\Omega$, that is, $XT_\Omega = T_\Omega(Y \otimes I_{\mathrm{env}})^{\mathsf{T}}$.

(a) If X_k and Y_k are mirrors for $k = 1, \ldots, n$, then $X_1 \cdots X_n$ and $Y_n \cdots Y_1$ are mirrors. Operators on different tensor factors commute, so $(X_1 X_2 \otimes I)\Omega = (X_1 \otimes Y_2 \otimes I)\Omega = (I \otimes Y_2 Y_1 \otimes I)\Omega$; induct. Linear combinations of mirror pairs are mirror pairs.

(b) If X and X' are both mirrors of Y, then $(X - X')P_A = 0$ by (N3). If Y and Y' are both mirrors of X, then $(Y - Y')P_B = 0$.

(c) Suppose X, Y are mirrors and X^*, Y^* are mirrors. This holds for self-adjoint mirrors, and for unitary mirrors because applying $X^* \otimes Y^*$ to their mirror identity gives $(I \otimes Y^* \otimes I)\Omega = (X^* \otimes I)\Omega$. Then $Xh_A = h_A X$ and $Yh_B = h_B Y$. For $Z \in B(\mathcal{H}_A)$,

$$\mathrm{Tr}(h_A Z X) = \langle (I \otimes Y^* \otimes I)\Omega, (Z \otimes I)\Omega \rangle = \langle (X^* \otimes I)\Omega, (Z \otimes I)\Omega \rangle = \mathrm{Tr}(h_A X Z).$$

Cyclicity of the trace, valid because h_A is trace class and X, Z are bounded, gives $\mathrm{Tr}((Xh_A - h_A X)Z) = 0$. Taking Z of rank one shows that the trace-class operator $Xh_A - h_A X$ is zero. Bob's statement is symmetric.

(d) An operator commuting with h_A commutes with P_A and with every spectral projection of h_A. In particular it maps each finite-dimensional positive eigenspace of h_A into itself.

(N5) Perfect matching. Let E be an Alice effect and F a Bob effect whose two disagreement events $E \otimes (I - F)$ and $(I - E) \otimes F$ have probability zero. As in Fact 2.2, the sum of these two probabilities is the expectation of

$$(E \otimes I - I \otimes F)^2 + (E - E^2) \otimes I + I \otimes (F - F^2),$$

a sum of three positive operators. Each expectation vanishes, so

$$(E \otimes I)\Omega = (I \otimes F \otimes I)\Omega, \qquad ((E - E^2)^{1/2} \otimes I)\Omega = 0, \qquad (I \otimes (F - F^2)^{1/2} \otimes I)\Omega = 0.$$

Thus E and F are mirrors. Next, E reduces P_A. Since $((I - P_A) \otimes I)\Omega = 0$,

$$((I - P_A)E \otimes I)\Omega = (I \otimes F \otimes I)((I - P_A) \otimes I)\Omega = 0,$$

so $(I - P_A)EP_A = 0$ by (N3), and self-adjointness gives $EP_A = P_A E P_A = P_A E$. Then (N3) gives $(E - E^2)P_A = 0$, so EP_A is a projection, and E commutes with h_A by (N4c). The same holds for F on Bob's side. We call EP_A a *synchronized projection* and mean this support-restricted operator in all local identities. Reduction of the support is proved before projectivity is used.

(N6) Compressions and joint measurements.

(a) If $0 \le q \le Z$ with Z a projection, then $q = ZqZ$. Indeed $0 \le (I-Z)q(I-Z) \le (I-Z)Z(I-Z) = 0$, so $q^{1/2}(I-Z) = 0$.

(b) *Projective compression.* If P is a projection, E an effect and PEP a projection, then E reduces $P\mathcal{H}$. For $u \in \mathrm{ran}(PEP)$, $\langle u, (I-E)u\rangle = 0$ and positivity give $Eu = u$. For $u \in P\mathcal{H}$ with $PEPu = 0$, $\langle u, Eu\rangle = 0$ gives $Eu = 0$. Thus E maps $P\mathcal{H}$ into itself, and self-adjointness gives reduction.

(c) *Only compressions matter.* Since $(P_A \otimes I)\Omega = \Omega = (I \otimes P_B \otimes I)\Omega$, every probability equals $\langle \Omega, (P_A E P_A \otimes P_B F P_B \otimes I)\Omega\rangle$. Identities may therefore be proved first for compressed effects. Changing effects outside the supports, while keeping them POVMs, changes no probability.

(d) *Sharp marginals.* Let (E_c) be a finite POVM indexed by tuples $c = (c_1, \dots, c_m)$ whose compressions to a reducing subspace $P\mathcal{H}$ have projections as coarse marginals $Q^k_s = \sum_{c:\, c_k = s} E_c$. By (a), each compressed E_c satisfies $E_c = Q^k_{c_k} E_c Q^k_{c_k}$, hence $E_c Q^k_s = \delta_{s,c_k} E_c$. Summing, $Q^1_{s_1} Q^2_{s_2} = \sum_{c_1 = s_1,\, c_2 = s_2} E_c$, which is self-adjoint, so the marginals commute. Iterating, $Q^1_{s_1} \cdots Q^m_{s_m}$ is the compressed effect of the outcome $(s_1, \dots, s_m)$, or zero if there is no such outcome. Each compressed fine effect is thus a product of commuting projections, hence a projection, and (b) makes the original fine effect reduce $P\mathcal{H}$.

(N7) Forbidden events. Let Q be an Alice effect and R a Bob effect whose compression $R_B = P_B R P_B$ is a synchronized projection with Alice mirror R_A. If the event $Q \otimes R$ has probability zero, then $QR_AP_A = 0$. If Q also reduces P_A, then $R_AQP_A = 0$ as well. Indeed $0 = \langle \Omega, (Q \otimes R_B \otimes I)\Omega\rangle = \|(Q^{1/2} \otimes R_B \otimes I)\Omega\|^2$, and $(Q^{1/2} \otimes R_B \otimes I)\Omega = (Q^{1/2} R_A \otimes I)\Omega$, so (N3) gives $Q^{1/2} R_A P_A = 0$. The second claim is the adjoint of the first, because P_A commutes with Q and R_A. No commutation with a density is used.

(N8) Canonical purifications for completeness. Let h be a faithful positive trace-class operator of trace one on $\mathcal{H}$. By (N1), $\mathcal{H}$ is separable and has an orthonormal basis (φ_i) with $h\varphi_i = \lambda_i \varphi_i$ and $\lambda_i > 0$. Put $\Omega_h = \sum_i \lambda_i^{1/2} \varphi_i \otimes \varphi_i$, and let X^{T} denote the transpose in (φ_i). Both marginals of Ω_h equal h. If X commutes with h, it commutes with $h^{1/2}$, and the rule of (N2) gives

$$(I \otimes X^{\mathsf{T}})\Omega_h = (X \otimes I)\Omega_h,$$

since both vectors correspond to $h^{1/2}X = Xh^{1/2}$. The map $X \mapsto X^{\mathsf{T}}$ is linear, reverses products, and preserves adjoints, positivity and projections; and X^{T} commutes with $h^{\mathsf{T}} = h$. Hence a centralizer projection X for Alice and X^{T} for Bob are perfectly matched: their disagreement probability is $\langle \Omega_h, ((X(I-X) + (I-X)X) \otimes I)\Omega_h\rangle = 0$. For centralizer effects Q of Alice and R^{T} of Bob, the joint probability is $\mathrm{Tr}(hQR)$, so an event with $QR = 0$ has probability zero.

4.1 The normal flagged magic-square bridge

The magic-square relations originate with Mermin and Peres [16, 19]. We use Mančinska's flagged magic-square (FMS) geometry [15]. We spell out the predicate and prove its normal extension because the published finite-dimensional statement does not suffice here. All FMS question and answer indices start at zero; frame indices below start at one.

4.1.1 Questions, calibration and exact bindings

Alice has $\mathsf{R}_0, \mathsf{R}_1, \mathsf{R}_2, \mathsf{T}_0, \mathsf{T}_1$ and Bob has $\mathsf{C}_0, \mathsf{C}_1, \mathsf{C}_2, \mathsf{B}_0, \mathsf{B}_1$. A row question has a flag answer f or a triple of product $+1$. A column question has f or a triple of product $+1, +1, -1$ according

to its column. The magic square is

$$\begin{pmatrix} XI & IX & XX \\ IZ & ZI & ZZ \\ XZ & ZX & YY \end{pmatrix}. \tag{6}$$

Here X, Y, Z denote the Pauli matrices, I the 2×2 identity, and adjacent letters their tensor product. The displayed matrices describe a canonical realization; the predicate only checks the corresponding signs. Select column-zero outcomes $c_0 = (+\,+\,+)$, $c_1 = (+\,-\,-)$ and row-zero outcomes $r_0 = (+\,+\,+)$, $r_1 = (+\,-\,-)$. Question T_x has answers $+, -$ and the three column-zero triples other than c_x; B_y analogously retains all row-zero triples other than r_y.

Table 3 and Figure 2 summarize the rules. On $\mathsf{R}_i, \mathsf{C}_j$, reject flag/nonflag mismatch and otherwise check the intersecting cell; two flags are accepted. On $\mathsf{T}_x, \mathsf{C}_0$, a sign is allowed only against f or c_x; a retained triple must equal the opposite triple. On $\mathsf{R}_0, \mathsf{B}_y$ use the analogous rule with r_y. On $\mathsf{T}_x, \mathsf{B}_y$, reject two signs $a, b \in \{+1, -1\}$ unless $ab = (-1)^{xy}$; all pairs with a nonsign answer are accepted. Other question pairs are accepted. For a conditional instance add an inactive answer $\perp$, accept two inactive answers, and reject exactly one inactive answer. Bind each noninactive event to its synchronized active projection and all row/column flags to the synchronized flag.

Let P_{mag} be the nonflag magic support. Its four column-zero projections $m_{00}, m_{01}, m_{10}, m_{11}$ correspond respectively to $(+\,+\,+), (+\,-\,-), (-\,+\,-), (-\,-\,+)$. On Alice, m_{00} is read from the retained $(+\,+\,+)$ answer of T_1 and the other three from T_0. On Bob all four come from C_0. Each of these chosen occurrences is matched to its opposite synchronized binary question. The positive nonflag ZI event is read from Alice's R_1 component 1 and Bob's C_1 component 1. The positive IX event comes from Alice's R_0 component 1 and Bob's C_1 component 0. Put

$$\mathfrak{x} = I - 2P_{\mathrm{mag}} + 2E_{ZI,+}, \quad \mathfrak{z} = I - 2P_{\mathrm{mag}} + 2E_{IX,+}, \quad t_{ab} = \mathfrak{x}^a \mathfrak{z}^b m_{00}.$$

These are reflections extended by identity outside P_{mag}, and $t_{ab}^* t_{ab} = m_{00}$, $t_{ab} t_{ab}^* = m_{ab}$.

Questions	Answers	Tested rule
$\mathsf{R}_i, \mathsf{C}_j$	flag or parity triple	flags match; nonflags agree at (i, j)
$\mathsf{T}_x, \mathsf{C}_0$	sign or retained triple	signs pair only with f, c_x; retained triples match
$\mathsf{R}_0, \mathsf{B}_y$	triple or sign	symmetric calibration at r_y
$\mathsf{T}_x, \mathsf{B}_y$	signs or triples	two signs satisfy $ab = (-1)^{xy}$
Conditional copies	also inactive $\perp$	activity matches; flags and events bind to named projections

Table 3: Flagged magic-square questions, answers and rules. Unspecified pairs are accepted.

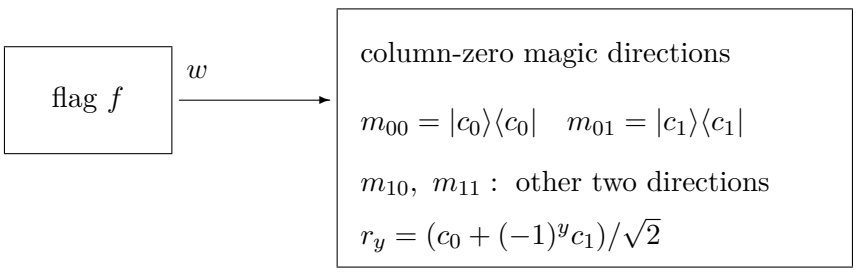


Figure 2: Selected column and row directions. The flag transports to c_0; m_{ab} denote projections.

4.1.2 Normal bridge theorem

Lemma 4.1. *A perfect normal conditional FMS strategy has on its active supports the decomposition*

$$\mathcal{H}_{\text{active}} = \mathcal{H}_f \oplus (\mathbb{C}^4 \otimes \mathcal{H}_{\text{mag}}), \qquad h = h_f \oplus (I_4 \otimes h_{\text{mag}}).$$

The calibrated transports w_A, w_B from the flag to the selected c_0 corner satisfy $w^*w = f$, $ww^* = m_{00}$ and $hw = 2wh$. On a purification their matched components satisfy

$$\Omega_{c_0c_0} = \sqrt{2}(w_A \otimes w_B \otimes I_{\text{env}})\Omega_{ff}. \tag{7}$$

The flag answer and its projection are both denoted f. The other three magic directions obey the transported versions of this equation.

Proof. First apply Section 2 to the activity, flags and checked cells. They reduce the original state supports and are sharp there. On the compressed nonflag space each compressed fine effect lies below its assigned sharp cell marginals. Thus it is their joint spectral projection. The projective-compression fact then makes the original fine effects reduce the support. Alice's row relations and Bob's column relations transfer through the dense coefficient range, in reversed product order; within a context the products commute.

In detail. Fix a purification Ω as in (N1). The predicate makes the following pairs of events perfectly matched in the sense of (N5): an inactive answer on one side and the inactive answer on the other; each noninactive event and the opposite synchronized activity projection; each row or column flag and the opposite synchronized flag; and, for every cell (i, j) and sign s, Alice's event "R_i is not a flag and its entry j equals s" and Bob's event "C_j is not a flag and its entry i equals s". The last pair is matched because the rule on $\mathsf{R}_i, \mathsf{C}_j$ rejects a flag against a nonflag and checks the intersecting cell otherwise. By (N5), each coarse effect in this list reduces its party's support, its compression is a projection commuting with that party's marginal density, and matched compressions are mirrors. Let a_A and f_A be Alice's activity and flag projections, put $N_A = a_A - f_A$, and let Q^s_{ij} be the compression of the cell event above, so that $Q^+_{ij} + Q^-_{ij} = N_A$. The cell observables $\mathsf{M}^A_{ij} = Q^+_{ij} - Q^-_{ij}$ are self-adjoint unitaries on $N_A P_A \mathcal{H}_A$. Each is a mirror of Bob's operator M^B_{ij}, built in the same way from C_j.

For a nonflag answer $t = (t_0, t_1, t_2)$ of R_i, the compressed effect lies below $Q^{t_j}_{ij}$ for each j, since the other nonflag effects of R_i with entry j equal to t_j are positive. By (N6d) it equals $Q^{t_0}_{i0}Q^{t_1}_{i1}Q^{t_2}_{i2}$, the three cell observables of row i commute, and the original fine effect reduces P_A. Summing over the triples of product $+1$ gives $\mathsf{M}^A_{i0}\mathsf{M}^A_{i1}\mathsf{M}^A_{i2} = N_A$ on $P_A\mathcal{H}_A$. The same argument for Bob's columns gives commuting $\mathsf{M}^B_{0j}, \mathsf{M}^B_{1j}, \mathsf{M}^B_{2j}$ with product $\epsilon_j N_B$, where $\epsilon_0 = \epsilon_1 = 1$ and $\epsilon_2 = -1$. By (N4a), $\mathsf{M}^A_{0j}\mathsf{M}^A_{1j}\mathsf{M}^A_{2j}$ is a mirror of $\mathsf{M}^B_{2j}\mathsf{M}^B_{1j}\mathsf{M}^B_{0j} = \epsilon_j N_B$, which is a mirror of $\epsilon_j N_A$. Then (N4b) gives $\mathsf{M}^A_{0j}\mathsf{M}^A_{1j}\mathsf{M}^A_{2j} = \epsilon_j N_A$ on $P_A\mathcal{H}_A$. In the same way $\mathsf{M}^A_{ij}\mathsf{M}^A_{i'j}$ and $\mathsf{M}^A_{i'j}\mathsf{M}^A_{ij}$ are mirrors of the equal operators $\mathsf{M}^B_{i'j}\mathsf{M}^B_{ij}$ and $\mathsf{M}^B_{ij}\mathsf{M}^B_{i'j}$, so they agree on $P_A\mathcal{H}_A$. The nine cell observables therefore satisfy every row and column relation of (6) as operator identities on $N_A P_A \mathcal{H}_A$. The symmetric argument gives the same for Bob on $N_B P_B \mathcal{H}_B$.

For completeness, write $a = \mathsf{M}_{00}, b = \mathsf{M}_{01}, c = \mathsf{M}_{10}, d = \mathsf{M}_{11}$ for the four top-left cell reflections of the square, where M_{ij} denotes the checked cell observable. Context relations give

$$[a, b] = [c, d] = [a, c] = [b, d] = 0, \qquad (ac)(bd) = -(ab)(cd).$$

Cancel the outside a, d to obtain $cb = -bc$. Commutation of ac and bd then gives $da = -ad$. The pairs (a, d) and (b, c) are anticommuting Pauli pairs commuting across pairs. Each pair exchanges its two eigenspaces unitarily. Hence they generate $M_4 \otimes I_{\mathcal{H}_{\text{mag}}}$. Marginal commutation gives the stated density form. The synchronized magic units identify its coefficient block as $I_4 \otimes T_{\text{mag}}$; write the flag coefficient block as T_f. On the cut $A \mid B\,\text{env}$ these maps

have dense range on Alice's corresponding supports, but need not be injective on their full domains.

In detail. Put $\mathcal{N}_A = N_A P_A \mathcal{H}_A$, and let $\mathcal{N}_\pm$ be the ± 1 eigenspaces of a, so that $\mathcal{N}_A = \mathcal{N}_+ \oplus \mathcal{N}_-$. Since $da = -ad$, d maps $\mathcal{N}_\pm$ into $\mathcal{N}_\mp$, and $d^2 = I$ makes $d|_{\mathcal{N}_+}$ unitary onto $\mathcal{N}_-$. This identifies $\mathcal{N}_A$ with $\mathbb{C}^2 \otimes \mathcal{N}_+$, with $a = Z \otimes I$ and $d = X \otimes I$. The operators b, c commute with a and d, hence with the algebra $M_2 \otimes I$ they generate, so $b = I \otimes b'$ and $c = I \otimes c'$ for anticommuting self-adjoint unitaries b', c'. Repeating the argument for c', b' gives $\mathcal{N}_+ = \mathbb{C}^2 \otimes \mathcal{H}_{\mathrm{mag}}$. No dimension count is used. The space $\mathcal{H}_{\mathrm{mag}}$ may be infinite-dimensional; it is separable because it lies in $P_A \mathcal{H}_A$.

For later use we record the identification through the units t_{ab}. The projection m_{00} is the joint $(+1,+1)$ eigenprojection of the column-zero cells $a = \mathsf{M}_{00}$ and $c = \mathsf{M}_{10}$, and m_{ab} is their joint $((-1)^a, (-1)^b)$ eigenprojection; this is the stated correspondence with $(++), (+-), (-+), (--)$. On $\mathcal{N}_A$ the reflection $\mathfrak{x}$ equals $d = \mathsf{M}_{11}$, which anticommutes with a and commutes with c, and $\mathfrak{z}$ equals $b = \mathsf{M}_{01}$, which commutes with a and anticommutes with c. Hence $t_{ab} = \mathfrak{x}^a \mathfrak{z}^b m_{00}$ maps $m_{00}\mathcal{N}_A$ isometrically onto $m_{ab}\mathcal{N}_A$, and

$$U_A : \mathbb{C}^4 \otimes \mathcal{H}^A_{\mathrm{mag}} \to \mathcal{N}_A, \qquad U_A(u_{ab} \otimes \xi) = t_{ab}\xi, \qquad \mathcal{H}^A_{\mathrm{mag}} = m_{00}\mathcal{N}_A,$$

is unitary. With $u_{ab} = |a\rangle \otimes |b\rangle$, these coordinates give $m_{ab} = |u_{ab}\rangle\langle u_{ab}| \otimes I$, $a = Z \otimes I$, $c = I \otimes Z$, $d = X \otimes I$ and $b = I \otimes X$. The operators $\mathfrak{x}, \mathfrak{z}, m_{00}$ are built from synchronized projections, so they commute with h_A by (N5), and so does each t_{ab}. Therefore $U_A^* h_A U_A = I_4 \otimes h^A_{\mathrm{mag}}$, where h^A_{mag} is the restriction of h_A to $\mathcal{H}^A_{\mathrm{mag}}$. Together with $[f_A, h_A] = 0$ this is the decomposition in the lemma, with $\mathcal{H}_f = f_A P_A \mathcal{H}_A$. Both blocks are faithful, because h_A is faithful on $P_A\mathcal{H}_A$ and they are its restrictions to reducing subspaces. The same construction on Bob's side gives U_B and h^B_{mag}.

Now the coefficient blocks. Flags and activity are matched, so $(f_A \otimes I)\Omega = (I \otimes f_B \otimes I)\Omega$ and $(N_A \otimes I)\Omega = (I \otimes N_B \otimes I)\Omega$. Put $\Omega_{ff} = (f_A \otimes f_B \otimes I)\Omega$ and $\Omega_{NN} = (N_A \otimes N_B \otimes I)\Omega$. The mixed components vanish; for example $(f_A \otimes N_B \otimes I)\Omega = (f_A N_A \otimes I)\Omega = 0$. Each of $\mathfrak{x}, \mathfrak{z}, m_{00}$ is a linear combination of matched events, hence a mirror of its Bob counterpart by (N4a); for $\mathfrak{x}$ the events are the activity, the flag and the positive ZI event, which the rule on $\mathsf{R}_1, \mathsf{C}_1$ matches. The reflections are self-adjoint, so (N4a) makes t^A_{ab} a mirror of $(t^B_{ab})^*$, and the matrix unit $t^A_{ab}(t^A_{a'b'})^*$ a mirror of $t^B_{a'b'}(t^B_{ab})^*$. In the coordinates $U_A \otimes U_B$ write $\Omega_{NN} = \sum_{k,l} u_k \otimes u_l \otimes \omega_{kl}$, where k, l range over the four labels ab and $\omega_{kl} \in \mathcal{H}^A_{\mathrm{mag}} \otimes \mathcal{H}^B_{\mathrm{mag}} \otimes \mathcal{H}_{\mathrm{env}}$. The mirror identity for the units $|u_k\rangle\langle u_l|$ and $|u_l\rangle\langle u_k|$ reads

$$\sum_{l'} u_k \otimes u_{l'} \otimes \omega_{ll'} = \sum_{k'} u_{k'} \otimes u_l \otimes \omega_{k'k},$$

so $\omega_{ll'} = 0$ for $l' \neq l$ and $\omega_{ll} = \omega_{kk}$. Hence

$$\Omega_{NN} = \sum_k u_k \otimes u_k \otimes \omega_{\mathrm{mag}}$$

for a single vector ω_{mag}, whose two marginals are h^A_{mag} and h^B_{mag}. In coefficient form, the coefficient operator of Ω_{NN} is $I_4 \otimes T_{\mathrm{mag}}$, where T_{mag} is that of ω_{mag}, and T_f is that of Ω_{ff}. Here $T_{\mathrm{mag}} : \mathcal{H}^B_{\mathrm{mag}} \otimes \mathcal{H}_{\mathrm{env}} \to \mathcal{H}^A_{\mathrm{mag}}$ and $T_f : \mathcal{H}^B_f \otimes \mathcal{H}_{\mathrm{env}} \to \mathcal{H}^A_f$ are the coefficient maps in the fixed product coordinates. They satisfy $T_{\mathrm{mag}} T^*_{\mathrm{mag}} = h^A_{\mathrm{mag}}$ and $T_f T_f^* = h^A_f$, and have dense range. They need not be injective. All cancellations below use (N3) on the appropriate original local support. Restricting a coefficient map to $(\ker T)^\perp$ and choosing Schmidt coordinates permits a positive amplitude operator, but is not needed here and need not preserve the Bob/environment factorization.

Calibration makes Alice's sign support $S^A_x = f + \Pi_{c_x}$ and Bob's $S^B_y = f + \Pi_{r_y}$. The selected directions have overlap $\langle c_x, r_y\rangle = \sigma_{xy}/\sqrt{2}$, where $\sigma_{xy} = (-1)^{xy}$. Their compressed

coefficient operator is

$$W_{xy} = \mathrm{diag}(T_f, \sigma_{xy} T_{\mathrm{mag}}/\sqrt{2}).$$

For compressed sign effects E_a, B_b, a forbidden pair gives $E_a^{1/2} W_{xy} (B_b^T)^{1/2} = 0$, and therefore $E_a W_{xy} B_b^T = 0$. Summing the two complete binary effects gives

$$E_a W_{xy} = W_{xy} B_{\sigma_{xy} a}^T = E_a W_{xy} B_{\sigma_{xy} a}^T.$$

The vector calculation below, followed by (N3) on each original local sign support, gives $B_b^2 = B_b$ and $E_a^2 = E_a$ there. Their zero extensions give projective compressions; thus the original sign effects reduce the state supports. Their differences $\mathcal{X}_x, \mathcal{Y}_y$ obey

$$\mathcal{X}_x W_{xy} = \sigma_{xy} W_{xy} \mathcal{Y}_y^T. \tag{8}$$

In detail. The rule on $\mathsf{T}_x, \mathsf{C}_0$ matches Alice's coarse sign event of T_x with Bob's event $\{f, c_x\}$ of C_0, and each retained triple of T_x with the same triple of C_0. Bob's C_0 projection for a triple is the joint projection of his column-zero cells, and by (N4a) its mirror is Alice's joint projection Π_c of her cells $\mathsf{M}_{00}^A, \mathsf{M}_{10}^A, \mathsf{M}_{20}^A$. Thus (N4b) gives $S_x^A = f_A + \Pi_{c_x}$ on $P_A \mathcal{H}_A$, and the retained effects of T_x are the other column-zero projections. Symmetrically, the rule on $\mathsf{R}_0, \mathsf{B}_y$ gives $S_y^B = f_B + \Pi_{r_y}$, where Π_{r_y} is Bob's joint projection of his row-zero cells. In the coordinates above, $c_0 = u_{00}$ and $c_1 = u_{01}$, while $r_y = (u_{00} + (-1)^y u_{01})/\sqrt{2}$, because row zero consists of $a = Z \otimes I$, $b = I \otimes X$ and ab. Hence $\langle c_x, r_y \rangle = \sigma_{xy}/\sqrt{2}$, and

$$\Omega_{xy} := (S_x^A \otimes S_y^B \otimes I)\Omega = \Omega_{ff} + \frac{\sigma_{xy}}{\sqrt{2}}\, c_x \otimes r_y \otimes \omega_{\mathrm{mag}}.$$

The maps $V_x^A(\xi \oplus \eta) = \xi + c_x \otimes \eta$ and $V_y^B(\xi' \oplus \eta') = \xi' + r_y \otimes \eta'$ identify $\mathcal{H}_f \oplus \mathcal{H}_{\mathrm{mag}}^A$ and $\mathcal{H}_f^B \oplus \mathcal{H}_{\mathrm{mag}}^B$ with the two sign supports. Under them Ω_{xy} becomes $\Omega_{ff} + \sigma_{xy}\omega_{\mathrm{mag}}/\sqrt{2}$, with the two terms in the flag-flag and the magic-magic blocks; its coefficient operator is W_{xy}. Its Alice marginal is $h_f \oplus h_{\mathrm{mag}}^A/2$ and its Bob marginal is $h_f^B \oplus h_{\mathrm{mag}}^B/2$; both are faithful.

Let $E'_a = P_A E_a P_A$ and $B'_b = P_B B_b P_B$ be the compressed sign effects of T_x and B_y. Then $0 \le E'_a \le S_x^A$ and $0 \le B'_b \le S_y^B$, so $E'_a = E'_a S_x^A$ and $B'_b = B'_b S_y^B$ by (N6a). If $ab \neq \sigma_{xy}$, the pair is rejected, so

$$0 = \langle \Omega, (E'_a \otimes B'_b \otimes I)\Omega \rangle = \|(E_a'^{1/2} \otimes B_b'^{1/2} \otimes I)\Omega\|^2,$$

whence $(E'_a \otimes B'_b \otimes I)\Omega = 0$. These are the vector forms of $E_a^{1/2} W_{xy} (B_b^T)^{1/2} = 0$ and $E_a W_{xy} B_b^T = 0$. Writing $\sigma = \sigma_{xy}$ and summing the complete binary effects,

$$(E'_a \otimes I)\Omega_{xy} = (E'_a \otimes S_y^B \otimes I)\Omega = (E'_a \otimes B'_{\sigma a} \otimes I)\Omega = (S_x^A \otimes B'_{\sigma a} \otimes I)\Omega = (I \otimes B'_{\sigma a} \otimes I)\Omega_{xy}.$$

Next, $(E'_a E'_{-a} \otimes I)\Omega_{xy} = (E'_a \otimes B'_{-\sigma a} \otimes I)\Omega_{xy} = (E'_a \otimes B'_{-\sigma a} \otimes I)\Omega = 0$. Since $E'_{-a} = S_x^A - E'_a$, this says $((E'_a - E_a'^2) \otimes I)\Omega_{xy} = 0$. The Alice marginal of Ω_{xy} is faithful on $S_x^A P_A \mathcal{H}_A$, so (N3), applied to Ω_{xy}, gives $E_a'^2 = E'_a$. This is where dense range enters. Bob's case is symmetric. By (N6b) the original sign effects reduce the supports. The differences $\mathcal{X}_x = E'_+ - E'_-$ and $\mathcal{Y}_y = B'_+ - B'_-$ are self-adjoint unitaries on $S_x^A P_A \mathcal{H}_A$ and $S_y^B P_B \mathcal{H}_B$, and

$$(\mathcal{X}_x \otimes I)\Omega_{xy} = \sigma_{xy}(I \otimes \mathcal{Y}_y \otimes I)\Omega_{xy}$$

is the vector form of (8).

Use the matrix units $t_{ab} t_{cd}^*$ to identify each selected magic direction with the same multiplicity space, and use $\sqrt{2}\Pi_{c_x}\Pi_{r_y}$ with its sign σ_{xy} for the selected row directions. Set $T_{\mathrm{cal}} = \mathrm{diag}(T_f, T_{\mathrm{mag}}/\sqrt{2})$ and $\Delta = \mathrm{diag}(I, -I)$. For $(x, y) = (0,0), (0,1), (1,0)$, Eq. (8) reads respectively $\mathcal{X}_0 T_{\mathrm{cal}} = T_{\mathrm{cal}} \mathcal{Y}_0^T$, $\mathcal{X}_0 T_{\mathrm{cal}} = T_{\mathrm{cal}} \mathcal{Y}_1^T$, and $\mathcal{X}_1 T_{\mathrm{cal}} = T_{\mathrm{cal}} \mathcal{Y}_0^T$. Writing these three

equations in vector form and applying (N3) on Alice's and Bob's respective local sign supports gives $\mathcal{X}_0 = \mathcal{X}_1 = \mathcal{X}$ and $\mathcal{Y}_0 = \mathcal{Y}_1 = \mathcal{Y}$. For $(1,1)$ it reads $\mathcal{X}\Delta T_{\rm cal} = -\Delta T_{\rm cal}\mathcal{Y}^T = -\Delta\mathcal{X}T_{\rm cal}$, so dense range gives $\mathcal{X}\Delta = -\Delta\mathcal{X}$. In blocks $\mathcal{X} = (\mathcal{X}_{ij})$, this says $\mathcal{X}_{11} = \mathcal{X}_{22} = 0$. Since $\mathcal{X} = \mathcal{X}^*$ and $\mathcal{X}^2 = I$, its remaining block $\beta = \mathcal{X}_{21}$ is unitary. Thus

$$\mathcal{X} = \begin{pmatrix} 0 & \beta^* \\ \beta & 0 \end{pmatrix}, \qquad \beta : \mathcal{H}_f \longrightarrow \mathcal{H}_{\rm mag} \text{ unitary.}$$

Self-adjointness and $\mathcal{X}T_{\rm cal} = T_{\rm cal}\mathcal{Y}^T$ imply $[\mathcal{X}, T_{\rm cal}T_{\rm cal}^*] = 0$. Here $h_f = T_f T_f^*$ and $h_{\rm mag} = T_{\rm mag}T_{\rm mag}^*$ are Alice's density blocks. The off-diagonal block gives $h_{\rm mag}\beta = 2\beta h_f$. The corresponding corner transport has the required ratio.

In detail. Put $\widehat{\mathcal{X}}_x = V_x^{A*}\mathcal{X}_x V_x^A$ and $\widehat{\mathcal{Y}}_y = V_y^{B*}\mathcal{Y}_y V_y^B$, and let $\Omega_{\rm cal} = \Omega_{ff} + \omega_{\rm mag}/\sqrt{2}$, whose coefficient operator is $T_{\rm cal}$. For $(x,y) \neq (1,1)$ the vector Ω_{xy} becomes $\Omega_{\rm cal}$, and for $(1,1)$ it becomes $(\Delta\otimes I)\Omega_{\rm cal} = (I\otimes\Delta\otimes I)\Omega_{\rm cal}$. The cases $(0,0)$ and $(1,0)$ give $((\widehat{\mathcal{X}}_0 - \widehat{\mathcal{X}}_1)\otimes I)\Omega_{\rm cal} = 0$, and the cases $(0,0)$ and $(0,1)$ give $(I\otimes(\widehat{\mathcal{Y}}_0 - \widehat{\mathcal{Y}}_1)\otimes I)\Omega_{\rm cal} = 0$. Both marginals of $\Omega_{\rm cal}$, namely $h_f \oplus h_{\rm mag}^A/2$ and $h_f^B \oplus h_{\rm mag}^B/2$, are faithful, so (N3) gives $\widehat{\mathcal{X}}_0 = \widehat{\mathcal{X}}_1 = \mathcal{X}$ and $\widehat{\mathcal{Y}}_0 = \widehat{\mathcal{Y}}_1 = \mathcal{Y}$. These are applications of (N3) to the two original local marginals; no injectivity on all of Bob tensor environment is assumed. The case $(1,1)$ then gives

$$(\mathcal{X}\Delta\otimes I)\Omega_{\rm cal} = -(I\otimes\mathcal{Y}\otimes I)(\Delta\otimes I)\Omega_{\rm cal} = -(\Delta\mathcal{X}\otimes I)\Omega_{\rm cal},$$

so $\mathcal{X}\Delta + \Delta\mathcal{X} = 0$ by (N3). A self-adjoint unitary anticommuting with $\Delta = \mathrm{diag}(I,-I)$ has zero diagonal blocks, and $\mathcal{X}^2 = I$ gives its lower block β the identities $\beta^*\beta = I_{\mathcal{H}_f}$ and $\beta\beta^* = I_{\mathcal{H}_{\rm mag}}$. The operators $\mathcal{X}, \mathcal{Y}$ are self-adjoint mirrors for $\Omega_{\rm cal}$, so (N4c) gives $[\mathcal{X}, h_f \oplus h_{\rm mag}^A/2] = 0$. In coefficient form this is the displayed commutator: from $\mathcal{X}T_{\rm cal} = T_{\rm cal}\mathcal{Y}^T$, the operator $\mathcal{X}T_{\rm cal}T_{\rm cal}^* = T_{\rm cal}\mathcal{Y}^T T_{\rm cal}^*$ is self-adjoint. Its off-diagonal block is $\beta h_f = \frac{1}{2}h_{\rm mag}^A\beta$. On Bob's side the same argument gives a unitary β_B with $h_{\rm mag}^B\beta_B = 2\beta_B h_f^B$.

The actual Alice extraction is $w_A = \Pi_{c_0}\mathcal{X}_0 f$. Bob's r_0 projection is the retained $(+++)$ answer of B_1, equivalently the sign support of B_0 minus its flag. His extraction to the same c_0 direction is

$$w_B = \sqrt{2}\Pi_{c_0}\Pi_{r_0}\mathcal{Y}_0 f.$$

The coefficient $\sqrt{2}$ occurs in extraction, not in the predicate. To see the phase in (7) directly, put $\Omega' = (S_0^A\otimes S_0^B)\Omega$. The zero sign events give $\mathcal{X}_0\Omega' = \mathcal{Y}_0\Omega'$. Project onto $c_0\otimes f$. The left side is $(w_A\otimes I)\Omega_{ff}$; the right side is

$$2^{-1/2}(I\otimes f\mathcal{Y}_0|r_0\rangle\langle c_0|)\Omega_{c_0c_0}.$$

Apply Bob's flag-to-r_0 block and then $|c_0\rangle\langle r_0| = \sqrt{2}\Pi_{c_0}\Pi_{r_0}$. This gives (7).

In detail. In the coordinates V_0^A, w_A is the map $\xi \mapsto c_0\otimes\beta\xi$ from $\mathcal{H}_f$ into $m_{00}P_A\mathcal{H}_A$. Thus $w_A^* w_A = f_A$ and $w_A w_A^* = m_{00}$ on the support, and $h_A w_A = 2w_A h_A$, because h_A acts as h_f on the flag and as $h_{\rm mag}^A$ on $c_0\otimes\mathcal{H}_{\rm mag}^A$. On Bob's side $\mathcal{Y}_0 f_B$ is $\xi' \mapsto r_0\otimes\beta_B\xi'$, and $\sqrt{2}\Pi_{c_0}\Pi_{r_0}$ restricts to the isometry $r_0\otimes\eta \mapsto c_0\otimes\eta$ from $\Pi_{r_0}P_B\mathcal{H}_B$ onto $m_{00}P_B\mathcal{H}_B$, since $\langle c_0, r_0\rangle = 2^{-1/2}$. So $w_B = c_0\otimes\beta_B$ has the same three properties. The components in $\mathcal{H}_{\rm mag}^A\otimes\mathcal{H}_f^B\otimes\mathcal{H}_{\rm env}$ of the two sides of $(\mathcal{X}\otimes I)\Omega_{\rm cal} = (I\otimes\mathcal{Y}\otimes I)\Omega_{\rm cal}$ give $(\beta\otimes I)\Omega_{ff} = (I\otimes\beta_B^*\otimes I)\omega_{\rm mag}/\sqrt{2}$, that is, $\omega_{\rm mag} = \sqrt{2}(\beta\otimes\beta_B\otimes I)\Omega_{ff}$. Since $\Omega_{c_0c_0} = c_0\otimes c_0\otimes\omega_{\rm mag}$, this is (7). The formula for Ω_{NN} also gives $\Omega_{m_{ab}m_{ab}} = (t_{ab}^A\otimes t_{ab}^B\otimes I)\Omega_{c_0c_0}$, and the components for two different magic directions vanish. These are the transported versions for the other three directions.

For a mixed state, first apply the pure-vector calculation on the cut $A \mid B\,\mathrm{env}$. The synchronized opposite effects reduce the Schmidt support, and the sign effects do so after their compressed sharpness is proved. Thus every compressed product used above is the restriction of an original Bob-only product tensored with $I_{\rm env}$. The resulting Alice identities hold on

her original marginal support. Repeat on $B \mid A\,\mathrm{env}$ for Bob's identities. The component calculation then gives Eq. (7) on the original purification, using $w_A \otimes w_B \otimes I_{\mathrm{env}}$. Tracing out the environment preserves the displayed component identity.

In detail. The calculation above already used the purification Ω on the cut $A \mid B\,\mathrm{env}$, with every Bob operator of the form $F \otimes I_{\mathrm{env}}$; the cut $B \mid A\,\mathrm{env}$ is the Bob form of (N3). The supports P_A, P_B are those of $\mathrm{Tr}_B\,\rho$ and $\mathrm{Tr}_A\,\rho$, so every identity holds on the original marginal supports. The partial trace over $\mathcal{H}_{\mathrm{env}}$ of the rank-one operators of the two sides of (7) gives $(m_{00} \otimes m_{00})\rho(m_{00} \otimes m_{00}) = 2(w_A \otimes w_B)(f \otimes f)\rho(f \otimes f)(w_A \otimes w_B)^*$. □

4.1.3 A shared question fixes transported labels

Lemma 4.2 (Label refinement)**.** *Retain the original FMS questions. For a synchronized projection P^{lab}, add Alice's $\mathsf{T}_0[P^{\mathrm{lab}}]$ with answers (a, z). Ignore z against the unchanged $\mathsf{B}_0, \mathsf{B}_1, \mathsf{C}_0$ and use the original predicates; match z to Bob's binary P^{lab}. Add the symmetric $\mathsf{B}_0[P^{\mathrm{lab}}]$ against unchanged $\mathsf{T}_0, \mathsf{T}_1, \mathsf{R}_0$. Also retain activity matching. In every perfect normal strategy the coarse signs equal the original signs and commute with P^{lab} on their state supports. Conversely commuting labels give these refinements. Distinct labels require no mutual commutation.*

Proof. Write P_A for the state support and R_A for the sign support. For an arbitrary new fine effect $Q_{a,z}$ first form $q_{a,z} = P_A Q_{a,z} P_A$. Calibration identifies its compressed retained effects and the coarse sign sum R_A. Let $\mathcal{X}'$ be its compressed sign difference. The unchanged opposite sign gives $\mathcal{X}' W_{0y} = W_{0y}\mathcal{Y}_y^T$, with exactly the same dense-range W_{0y} as for $\mathcal{X}_0$. Hence $\mathcal{X}' = \mathcal{X}_0$. Its coarse effects are $(R_A \pm \mathcal{X}_0)/2$, hence projections. Projective compression now makes the original coarse effects reduce P_A.

In detail. Let $q_a = \sum_z q_{a,z}$ be the compressed coarse effects of $\mathsf{T}_0[P^{\mathrm{lab}}]$. Against C_0 the original rule is applied to a, so the matching argument in the proof of Lemma 4.1 gives, by (N5) and (N4b), $q_t = \Pi_t$ for each retained triple t and $q_+ + q_- = R_A = f_A + \Pi_{c_0}$. Against $\mathsf{B}_0, \mathsf{B}_1$ the forbidden sign pairs are the original ones. The calculation in that proof which gave $(E'_a \otimes I)\Omega_{0y} = (I \otimes B'_a \otimes I)\Omega_{0y}$ and $E'^2_a = E'_a$ used only these pairs and the projection R_A, so it applies verbatim to $q_\pm$. With $\mathcal{X}' = q_+ - q_-$ it gives

$$(\mathcal{X}' \otimes I)\Omega_{00} = (I \otimes \mathcal{Y}_0 \otimes I)\Omega_{00} = (\mathcal{X}_0 \otimes I)\Omega_{00},$$

the vector form of $\mathcal{X}' W_{00} = W_{00}\mathcal{Y}_0^T$. The Alice marginal of Ω_{00} is faithful on $R_A P_A \mathcal{H}_A$, so (N3) gives $\mathcal{X}' = \mathcal{X}_0$ and $q_\pm = (R_A \pm \mathcal{X}_0)/2$.

Label matching gives $P_A(\sum_a Q_{a,1})P_A = P_A^{\mathrm{lab}}$. Positivity gives $0 \le q_{a,1} \le P_A^{\mathrm{lab}}$ and $0 \le q_{a,0} \le I - P_A^{\mathrm{lab}}$. Thus the compressed coarse effects commute with P_A^{lab}, and so does $\mathcal{X}_0$. The symmetric argument applies on Bob. Conversely use $E_a P^{\mathrm{lab}}, E_a(I - P^{\mathrm{lab}})$ when the original coarse effects commute with P^{lab}. A separate duplicated question for each label always returns the same original sign. This proves the final assertion.

In detail. The z-marginal $\sum_a Q_{a,1}$ is perfectly matched with Bob's binary question for P^{lab}. By (N5) its compression is a projection and a mirror of Bob's synchronized projection, whose Alice mirror is P_A^{lab}; (N4b) gives equality. For every a, positivity gives $0 \le q_{a,1} \le P_A^{\mathrm{lab}}$ and $0 \le q_{a,0} \le P_A - P_A^{\mathrm{lab}}$. By (N6a), $q_{a,1} = P_A^{\mathrm{lab}} q_{a,1} P_A^{\mathrm{lab}}$ and $q_{a,0} = (P_A - P_A^{\mathrm{lab}})q_{a,0}(P_A - P_A^{\mathrm{lab}})$, so $P_A^{\mathrm{lab}} q_a = q_{a,1} = q_a P_A^{\mathrm{lab}}$. All of these are identities on $P_A \mathcal{H}_A$. For the converse, if the original effects E_a commute with P^{lab}, then $E_a P^{\mathrm{lab}}$ and $E_a(I - P^{\mathrm{lab}})$ are positive. Summing over z gives E_a, and summing the $z = 1$ effects over a gives P^{lab}. Every probability involving only a is an original probability, and the z-test is the original perfect matching of P^{lab}. By (N6c) one may first replace P^{lab} by its support-restricted projection. No two different labels are measured jointly, so no commutation between them enters either direction. □

If a label is block diagonal with flag block P_f^{lab} and identical magic blocks P_m^{lab}, the commutator says $\beta P_f^{\mathrm{lab}} = P_m^{\mathrm{lab}}\beta$. In particular, for a graph projection of a block unitary it certifies transport of that same unitary by the calibrated w. It does not supply a fresh bridge with an independent gauge. Indeed, with respect to $R_A = f \oplus \Pi_{c_0}$ the commutator $[\mathcal{X}_0, P^{\mathrm{lab}}]$ has the (c_0, f) block $\beta P_f^{\mathrm{lab}} - P_m^{\mathrm{lab}}\beta$.

4.2 Rational projection equations that certify products

All projections in this section are synchronized. We give the finite rules used in the compiler. The projection-partition viewpoint is standard [9]; the formulas here provide the needed actual multiplication identities. See also the graph-projection encoding in [1, Section 4].

4.2.1 Graphs and triangles

Let F_i, F_j be orthogonal, $S = F_i + F_j$, and introduce projections $G, H, G^\perp, H^\perp$ with

$$G + G^\perp = S, \quad H + H^\perp = S, \quad 5H + S = 3F_i + 4G. \tag{9}$$

On S, put $\Delta_{ij} = F_i - F_j$, $\Xi_{ij} = 2G - S$. These are reflections, and $2H - S = (3\Delta_{ij} + 4\Xi_{ij})/5$. Squaring forces $\Delta_{ij}\Xi_{ij} + \Xi_{ij}\Delta_{ij} = 0$. Thus

$$\Xi_{ij} = \begin{pmatrix} 0 & u_{ij} \\ u_{ij}^* & 0 \end{pmatrix}, \qquad u_{ij}^* u_{ij} = F_j, \quad u_{ij} u_{ij}^* = F_i.$$

The graph blocks are certified, rather than assumed. Conversely any such unitary corner map gives the four projections in (9).

For a partition $F_1 + F_2 + F_3 = I$ impose the graph tests on all three pairs and add a projection $P_\triangle$ with

$$3P_\triangle = 2(G_{12} + G_{23} + G_{13}) - I. \tag{10}$$

Its block matrix has diagonal $I/3$ and off-diagonal $u_{ij}/3$. The $(1,3)$ block of $P_\triangle^2 = P_\triangle$ gives $2u_{13} + u_{12}u_{23} = 3u_{13}$, hence $u_{13} = u_{12}u_{23}$. Conversely this product identity makes the entire matrix a projection.

Start with a reference triangle for identity. Its scalar matrix algebra is called the *frame*. It gives scalar matrix units and $\mathcal{H} = \mathbb{C}^3 \otimes \mathcal{H}_{\mathrm{base}}$. For every unitary label U, use graphs in all three positions. Triangles $(\mathsf{U}, 1, \mathsf{U})$ and $(1, \mathsf{U}, \mathsf{U})$ identify the placements. A triangle $(\mathsf{U}, \mathsf{V}, \mathsf{W})$ then certifies $\mathsf{W} = \mathsf{UV}$; a triangle with product identity certifies inverse labels. Build each word from its proper prefix and final token. Finally a real rational equation $\sum_w q_w \mathsf{U}_w = 0$ is precisely

$$2\sum_w q_w G_{12}(\mathsf{U}_w) - \Big(\sum_w q_w\Big)(F_1 + F_2) = 0. \tag{11}$$

Its two off-diagonal blocks are the equation and its adjoint. All products therefore refer to actual common operators. Synchronization makes the extracted unitaries commute with the base marginal density.

4.2.2 Binding prescribed parameters

A prescribed synchronized projection P^{par} is linked to the frame by projections $P^{\mathrm{par}}F_i, NF_i$ with

$$P^{\mathrm{par}}F_i + NF_i = F_i, \quad \sum_i P^{\mathrm{par}}F_i = P^{\mathrm{par}}, \quad \sum_i NF_i = I - P^{\mathrm{par}}.$$

These symbols are names for projections before the equations identify their products; the names NF_i denote complements of $P^{\mathrm{par}}F_i$ inside F_i. For every pair ij, introduce $B_{11}, B_{10}, B_{01}, B_{00}$ and impose

$$B_{11}+B_{10}=P^{\mathrm{par}}F_i+P^{\mathrm{par}}F_j, \qquad B_{00}+B_{01}=NF_i+NF_j,$$
$$B_{11}+B_{01}=G_{ij}(1), \qquad G_{ij}(s_{\mathrm{par}})=B_{11}+B_{00}.$$

Projection sums give the indicated joint partitions. The symbol s_{par} is a unitary parameter label. These equations force P^{par} to commute with the frame and identify the extracted s_{par} with $2e-I$, where $P^{\mathrm{par}}=I_3\otimes e$. In particular these parameters are the specified FMS events, not independent replacements. The four-bridge assembly below uses twenty-five such parameters.

These rules are a deterministic finite compiler for the finite word list. Cancellation of adjacent inverse tokens is performed before taking proper prefixes. All coefficients stay rational. Conversely a centralizer unitary assignment with the prescribed parameter projections gives every graph, complement, H, triangle and joint projection simultaneously by these formulas.

Scope in infinite dimensions. All projections in this section are support-restricted synchronized projections on Alice's support $P_A\mathcal{H}_A$, as in (N5), and every displayed equation holds there as an operator identity by Lemma 4.3. The derivations are block-matrix calculations for orthogonal decompositions of $P_A\mathcal{H}_A$. Projections whose sum is a projection are mutually orthogonal, and a self-adjoint unitary anticommuting with $\mathrm{diag}(I,-I)$ has the off-diagonal form used for Ξ_{ij}. No trace, rank or dimension is used. The frame identification is the unitary

$$\mathbb{C}^3\otimes\mathcal{H}_{\mathrm{base}}\to P_A\mathcal{H}_A, \qquad e_i\otimes\xi\mapsto u_{i1}\xi, \qquad \mathcal{H}_{\mathrm{base}}=F_1P_A\mathcal{H}_A,$$

where $u_{11}=F_1$, $u_{21}=u_{12}^*$, $u_{31}=u_{13}^*$, and $u_{ij}=F_i(2G_{ij}(1)-F_i-F_j)F_j$. The frame projections and the maps u_{ij} commute with h_A by (N5), so $h_A=I_3\otimes h_{\mathrm{base}}$ in these coordinates. An extracted label U is defined by $F_1(2G_{12}(\mathsf{U})-F_1-F_2)F_2=e_{12}\otimes\mathsf{U}$, and it commutes with h_{base}.

The binding equations are solved in the same way. Write $\pi_i=P^{\mathrm{par}}F_i$ and $\nu_i=NF_i$. These are projections below F_i with $\pi_i+\nu_i=F_i$, and $\sum_i\pi_i=P^{\mathrm{par}}$, so $P^{\mathrm{par}}F_i=\pi_i=F_iP^{\mathrm{par}}$ and P^{par} commutes with every F_i. For a pair ij the four projections B_{ab} sum to $S=F_i+F_j$ and are therefore mutually orthogonal. Both $\pi=\pi_i+\pi_j$ and $G_{ij}(1)$ are sums of them, so they commute and $\pi G_{ij}(1)=B_{11}$. In frame coordinates write $\pi_i=e_{ii}\otimes p_i$. Commutation of π with $\Xi_{ij}(1)=(e_{ij}+e_{ji})\otimes I$ gives $p_i=p_j$; over all pairs, $P^{\mathrm{par}}=I_3\otimes e$. Finally $B_{00}=(S-\pi)(S-G_{ij}(1))$, so

$$G_{ij}(s_{\mathrm{par}})=\pi G_{ij}(1)+(S-\pi)(S-G_{ij}(1))=\tfrac{1}{2}\big(S+(e_{ij}+e_{ji})\otimes(2e-I)\big),$$

which identifies s_{par} with $2e-I$.

4.3 Rational rows as Boolean winning conditions

We implement each rational row by a subspace orthogonality test. This is the general rational-row theorem of [8, Theorem B], restated here to specify the extension and its quantitative loss.

Lemma 4.3 (Boolean rational rows)**.** *Let a finite strategy specification name finitely many synchronized binary projections and finitely many rational linear identities among them. There is a finite extension by Boolean tests with the following properties.*

Soundness: *in every perfect normal spatial strategy for the extended tests, the identities hold as operator identities for the named support-restricted projections on each local state support.*

Completeness: *any normal spatial strategy with these synchronized projections, satisfying the identities on its supports, extends to satisfy all the added tests perfectly after adjoining a finite maximally entangled state. The old measurements act trivially on the new factor. The rest of the old strategy, including nonsynchronous questions, is retained; if its old tests were perfect, the whole extension is perfect.*

4.3.1 A certified scalar matrix algebra

Choose $d_{\max} = 2^{2r}$ large enough for every row construction below. Use r synchronized magic squares, binding each context's cell marginals to synchronized binary cells. The Pauli-pair calculation in Lemma 4.1 gives two anticommuting pairs per square. Four-answer joint questions for all pairs of axes from distinct squares force cross-square commutation. Hence their algebra is $M_{d_{\max}} \otimes I$ on the support. Synchronization makes the marginal $I_{d_{\max}}/d_{\max} \otimes h$.

Add joint questions between every named old projection and every scalar axis. Their sharp marginals force the old projections into the commutant: $E_i = I_{d_{\max}} \otimes e_i$. This step concerns only synchronized projections, never an FMS sign. Add global Z and global X measurements with all bit outcomes, binding each bit marginal to its axis. They are the actual joint spectral measurements.

In detail. The Pauli-pair calculation is the one beginning "For completeness, write $a = \mathsf{M}_{00}$" in the proof of Lemma 4.1, without flags. A four-answer joint question for axes σ, σ' of distinct squares has its two binary marginals bound to synchronized cells, so the axes commute on $P_A\mathcal{H}_A$ by (N6d). Iterating the eigenspace argument of that proof over the $2r$ mutually commuting anticommuting pairs identifies $P_A\mathcal{H}_A$ with $\mathbb{C}^{d_{\max}} \otimes \mathcal{H}'$, the axes generating $M_{d_{\max}} \otimes I$; the space $\mathcal{H}'$ may be infinite-dimensional. The axes commute with h_A by (N5), so h_A lies in $(M_{d_{\max}} \otimes I)' = I \otimes B(\mathcal{H}')$, which is the stated form. A joint question between a named projection E_i and an axis makes them commute by (N6d). The axes generate $M_{d_{\max}} \otimes I$ as an algebra, so E_i lies in its commutant, and $E_i = I_{d_{\max}} \otimes e_i$ for a projection e_i on $\mathcal{H}'$. The global Z and X measurements have sharp bit marginals, so by (N6d) they are the joint spectral measures of the commuting Z axes and of the commuting X axes.

For a row dimension $d = 2^n \le d_{\max}$ use the first n qubits; the rest are spectators. For each nonzero $j \in \mathbb{F}_2^n$, choose a basis $\ell_1, \dots, \ell_{n-1}$ of $j^\perp$. Add a joint context for $X^j, Z^{\ell_1}, \dots, Z^{\ell_{n-1}}$, binding its marginals to the corresponding global parity events. These commute because $\ell \cdot j = 0$. The zero-Z-syndrome outcomes with X^j sign $\pm$ are

$$R_j^{\pm} = \tfrac{1}{2}(|0\rangle \pm |j\rangle)(\langle 0| \pm \langle j|) \otimes I_{\text{spectator}}.$$

Write $|+_d\rangle = d^{-1/2} \sum_{j=0}^{d-1} |j\rangle$. The reference $R_d = |+_d\rangle\langle +_d| \otimes I_{\text{spectator}}$ is the event that the first n global X bits are positive. It includes all spectator outcomes.

4.3.2 Signed reflections and column graphs

For one row $S = \alpha_0 I + \sum_i \alpha_i e_i$, let L clear all coefficient denominators and set

$$s_i = 2e_i - I, \quad c_i = L\alpha_i, \quad c_0 = 2L\alpha_0 + \sum_i L\alpha_i.$$

Then $2LS = c_0 I + \sum_i c_i s_i$. Expand this into signed reflections with integer repetitions, add a canceling $I, -I$ pair with $V_0 = I$, and pad by further such pairs to a power of four d. The list has even length because its coefficient sum is even. Thus

$$\sum_{j=0}^{d-1} V_j = 2LS, \qquad V_0 = I, \qquad V_j \in \{\pm I,\ \pm(2e_i - I)\}. \tag{12}$$

For each $j > 0$ use a six-answer joint question of e_i and the three-way scalar measurement $R_j^+, R_j^-, I - R_j^+ - R_j^-$. For $V_j = \pm I$, bind its binary bit to a synchronized deterministic anchor: its 1-projection is I and its 0-projection is zero. Matching both marginals identifies all joint effects. Appropriate coarse events are the two-coordinate graph projections

$$G_j^{\pm} = \frac{1}{2}\begin{pmatrix} I & \pm V_j \\ \pm V_j & I \end{pmatrix}$$

on coordinates $0, j$, zero elsewhere. A negative reflection sign exchanges the two coarse events.

Add a synchronized d-answer question with projections Π_t, indexed by $t \in \mathbb{F}_2^n$. Put $\chi_t(j) = (-1)^{t \cdot j}$. Forbid Π_t together with G_j^- when $\chi_t(j) = 1$, or G_j^+ when $\chi_t(j) = -1$. The forbidden-event support rule gives actual orthogonality. A vector in $\operatorname{ran}\Pi_t$ therefore has coordinates $v_j = \chi_t(j) V_j v_0$.

Define

$$\mathcal{I}_t \xi = d^{-1/2} (\chi_t(j) V_j \xi)_{j=0}^{d-1}, \qquad \mathcal{P}_t = \mathcal{I}_t \mathcal{I}_t^*.$$

Unitarity and character orthogonality give $\mathcal{I}_t^* \mathcal{I}_u = \delta_{tu} I$ and $\sum_t \mathcal{P}_t = I$. Neither equation requires different V_j to commute. The kernel conditions give $\Pi_t \le \mathcal{P}_t$. Since both families sum to identity, $\Pi_t = \mathcal{P}_t$ for every t.

Finally forbid Π_0 jointly with the opposite R_d. Then $\Pi_0 R_d = 0$, equivalently $R_d \mathcal{I}_0 = 0$. The column overlap is

$$\mathcal{I}_0^* R_d \mathcal{I}_0 = d^{-2} \Big(\sum_j V_j\Big)^* \Big(\sum_j V_j\Big).$$

It vanishes exactly when $S = 0$. This proves soundness on arbitrary normal supports.

In detail. The forbidden pairs are handled by (N7). Take Q to be the coarse event $G_j^{\mp}$ of the party that asks it and R the opposite synchronized Π_t. Both reduce the support, so $G_j^{\mp} \Pi_t = \Pi_t G_j^{\mp} = 0$ on that party's support. The events $G_j^{\pm}$ and Π_t are products of mirrored projections, so (N4a) and (N3) transfer this identity to the other party. In particular it holds on $P_A \mathcal{H}_A$. Write a vector of $P_A \mathcal{H}_A = \mathbb{C}^d \otimes \mathbb{C}^{d_{\max}/d} \otimes \mathcal{H}'$ as $(v_j)_{j \in \mathbb{F}_2^n}$, and regard each V_j as a support-restricted reflection, a self-adjoint unitary on $\mathbb{C}^{d_{\max}/d} \otimes \mathcal{H}'$. If $\chi_t(j) = 1$, the equation $G_j^- v = 0$ says $v_j = V_j v_0$. If $\chi_t(j) = -1$, the equation $G_j^+ v = 0$ says $v_j = -V_j v_0$. Thus $\operatorname{ran}\Pi_t \subseteq \operatorname{ran}\mathcal{I}_t$, that is, $\Pi_t \le \mathcal{P}_t$. The two families are complete, so $\sum_t (\mathcal{P}_t - \Pi_t) = 0$ with positive terms, and $\Pi_t = \mathcal{P}_t$. The forbidden pair (Π_0, R_d) gives $R_d \mathcal{P}_0 = 0$ by (N7), and $R_d \mathcal{I}_0 = R_d \mathcal{P}_0 \mathcal{I}_0 = 0$. Finally $(\langle +_d| \otimes I)\mathcal{I}_0 = d^{-1} \sum_j V_j = (2L/d) S$, so

$$\|R_d \mathcal{I}_0 \xi\|^2 = \frac{4L^2}{d^2} \|S\xi\|^2$$

for every ξ in the support. Hence $R_d \mathcal{I}_0 = 0$ holds exactly when S vanishes on the support. No inverse of S or of a density is used.

For completeness append $\Phi_{d_{\max}}$, use the scalar Paulis and the actual joint products, and define the column partition by the $\mathcal{I}_t$. Synchronized original projections commute with the marginal and transfer across a purification. Products transfer in reversed order. Define Bob's column effects by those transferred products, equivalently by transpose in canonical Schmidt coordinates. This order matters for noncommuting e_i. All column and synchrony tests hold.

The row-zero test holds precisely because $S = 0$. These constructions act trivially on every old measurement's new factor and prove completeness, including for mixed original states.

In detail. Let the old strategy have purification Ω and supports P_A, P_B. By (N6c) we may replace every named synchronized effect by its support-restricted projection, completed by a fixed outcome off the support; no probability changes. Adjoin $\Phi_{d_{\max}}$, so that the new purification is $\Omega' = \Omega \otimes \Phi_{d_{\max}}$ and the new state is again normal. Alice's new effects are the displayed operators: scalar Paulis, joint products $E_i\sigma$ and $E_i R_j^{\pm}$ of commuting projections, and the column projections $\mathcal{P}_t = \mathcal{I}_t\mathcal{I}_t^*$, all defined on the support and completed by a fixed outcome off it. For an Alice operator $\sum_k z_k \otimes p_k(e)$, with scalar factors z_k and products $p_k(e)$ of named projections, Bob uses $\sum_k z_k^{\mathsf{T}} \otimes p_k^{\mathrm{rev}}(e^B)$, where $p_k^{\mathrm{rev}}(e^B)$ is the reversed product of his matched projections. By (N4a) and $(z \otimes I)\Phi_{d_{\max}} = (I \otimes z^{\mathsf{T}})\Phi_{d_{\max}}$, it is a mirror of the Alice operator. This assignment commutes with adjoints, so the counterpart Y of an Alice projection P is self-adjoint. Moreover $(I \otimes (Y^2 - Y) \otimes I)\Omega' = ((P^2 - P) \otimes I)\Omega' = 0$, so Y is a projection on Bob's support by (N3), and the counterparts of the column projections sum to the identity. Every matching test therefore has disagreement probability zero. Every forbidden pair has probability zero because the corresponding Alice product vanishes, and the row-zero pair does so because $S = 0$ on the support. The old measurements act as $E \otimes I_{d_{\max}}$, so all old tests, including nonsynchronous ones, keep their probabilities.

Write h_{old} for the trace-one marginal before the row ancilla is appended (including the qutrit frame when present). For the honest column construction, without assuming $S = 0$, the sole row-zero event has probability

$$p(\Pi_0, R_d) = \frac{4L^2}{d^3} \operatorname{Tr}(h_{\mathrm{old}} S^2). \tag{13}$$

Indeed the scalar density is I_d/d, the column isometry is normalized by $d^{-1/2}$, and the reference vector by $d^{-1/2}$. The overlap computed above supplies $(2LS)^*(2LS)$. Spectator factors have normalized trace one. All column-graph and matching tests remain exact even when the row is nonzero. This formula is [8, Eq. (32)] in the general row setting.

In detail. Bob's reference effect is the transpose of R_d on the scalar factor and is a mirror of R_d. The probability is therefore $\|(\mathcal{P}_0 \otimes R_d^{\mathsf{T}})\Omega'\|^2 = \operatorname{Tr}(h' R_d \mathcal{P}_0 R_d)$, where $h' = (I_d/d) \otimes (d/d_{\max}) I_{\mathrm{spectator}} \otimes h_{\mathrm{old}}$. Since $(\langle +_d| \otimes I)\mathcal{I}_0 = (2L/d)S$, we have $R_d\mathcal{P}_0 R_d = |+_d\rangle\langle +_d| \otimes I_{\mathrm{spectator}} \otimes (4L^2/d^2)S^2$. The trace against h' is (13): the factor $1/d$ is $\langle +_d| I_d/d |+_d\rangle$, and the spectator trace is one. The calculation uses only that the V_j are self-adjoint unitaries, so it holds for every row, zero or not.

4.4 The admissible presentations

Definition (admissible presentation; normal realization). An admissible presentation is a finite rational list with four contraction symbols A_α ($\alpha \in \{0,1\}^2$), a centralizer projection label $P_\circ$ for the root, and finitely many centralizer labels, each typed as a unitary, projection, positive contraction ($0 \le T \le I$), or general contraction ($\|T\| \le 1$). Its application equations (the identities enforced by the final row tests) are rational $*$-polynomials in the centralizer labels, the creation and range equations

$$A_\alpha^* A_\beta = \delta_{\alpha\beta} I, \qquad \sum_\alpha A_\alpha A_\alpha^* = I - P_\circ,$$

and endpoint equations $LA_\alpha = A_\beta B$ or $LA_\alpha = 0$. Finite sums $\sum_\nu B_\nu A_{\alpha_\nu} C_\nu = 0$ are allowed. Here L, B, B_ν, C_ν are centralizer polynomial expressions. Apart from the displayed creation and range equations, no monomial contains more than one creation symbol. Every label used on a left endpoint is included in the transported unitary list Λ defined in the proof of Lemma 4.4.

A normal realization consists of a nonzero Hilbert space $\mathcal{H}_0$, a faithful positive trace-class operator h_0 with $\operatorname{Tr} h_0 = 1$, and bounded operators of the declared types on $\mathcal{H}_0$. Every centralizer label, including $P_\circ$, commutes with h_0. All application equations hold as operator identities on $\mathcal{H}_0$, and

$$h_0 A_\alpha = A_\alpha h_0/16.$$

Thus all identities are imposed on the state support, where the density is faithful. No positive lower spectral bound is required for a positive contraction. Finite entropy of a realization means $S(h_0) < \infty$.

Lemma 4.4 (Four-bridge compiler)**.** *Every admissible presentation effectively yields a finite rational Boolean game. Every perfect normal spatial strategy yields a normal realization in the sense just defined, on the nonzero source corner with faithful trace-one density $h_0 = 121 e_0 h e_0$. Every such normal realization extends to a perfect normal spatial strategy after adjoining the scaffold and finite ancillary factors. Finite marginal entropy is preserved by the completeness construction.*

Proof. We give the assembly and both directions.

Corners and parameters. Use seventeen orthogonal synchronized corners

$$e_0; \quad m_{g,00}, m_{g,01}, m_{g,10}, m_{g,11} \quad (1 \le g \le 4).$$

Put $e_g = m_{g,00}$ and $M_g = \sum_{ab} m_{g,ab}$, and impose the cover $e_0 + \sum_g M_g = I$. Conditional bridge g has flag e_{g-1}, magic support M_g and activity $e_{g-1} + M_g$. Use all event bindings described above in both orientations, including the flag/four-magic/inactive partition. Bind to the qutrit frame the seventeen corners and the eight positive ZI, IX events. These are the twenty-five frame-bound parameters. Introduce eight axis labels $\mathfrak{x}_g, \mathfrak{z}_g$ with the explicit equations

$$\mathfrak{x}_g = I - 2M_g + 2E_{g,ZI,+}, \qquad \mathfrak{z}_g = I - 2M_g + 2E_{g,IX,+}.$$

The calibrated partial isometries satisfy $hw_g = 2w_g h$. Thus

$$v = w_4 w_3 w_2 w_1, \quad v^* v = e_0, \quad v v^* = e_4, \quad hv = 16vh. \tag{14}$$

Trace cyclicity gives four magic masses twice the preceding flag mass in each bridge. The cover fixes

$$\operatorname{Tr}(he_0) = 1/121, \qquad 121 = 1 + 4(2 + 4 + 8 + 16). \tag{15}$$

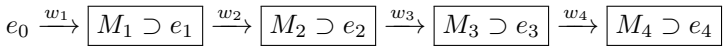
In particular the source corner is nonzero.

In detail. Let a perfect normal spatial strategy be given, with purification and supports as in (N1). By (N5) every corner, frame and graph question yields a synchronized projection on $P_A \mathcal{H}_A$ commuting with h_A, and by Section 4.2 and Lemma 4.3 every submitted equation, including the cover, holds on $P_A \mathcal{H}_A$. Projections with sum I are mutually orthogonal, so the seventeen corners partition $P_A \mathcal{H}_A$. Bridge g is a perfect conditional FMS strategy with activity $e_{g-1} + M_g$, flag e_{g-1} and magic support M_g, whose magic directions are the bound corners $m_{g,ab}$. Lemma 4.1 gives partial isometries w_g with $w_g^* w_g = e_{g-1}$, $w_g w_g^* = e_g$ and $h_A w_g = 2 w_g h_A$ on the support. Hence $v^* v = e_0$ and $v v^* = e_4$, and moving h_A through the four factors gives $h_A v = 16 v h_A$. For the masses, h_A is trace class and w_g is bounded, so cyclicity gives

$$\operatorname{Tr}(h_A e_g) = \operatorname{Tr}(h_A w_g w_g^*) = \operatorname{Tr}(w_g^* h_A w_g) = 2 \operatorname{Tr}(w_g^* w_g h_A) = 2 \operatorname{Tr}(h_A e_{g-1}).$$

Each t_{ab} of bridge g commutes with h_A, so $\operatorname{Tr}(h_A m_{g,ab}) = \operatorname{Tr}(h_A e_g)$. Summing over the cover, with $\operatorname{Tr} h_A = 1$, gives (15).

Figure 3 displays the four-bridge chain.



corner	e_0	e_1	e_2	e_3	e_4
$121\,\mathrm{Tr}(he_g)$	1	2	4	8	16
$121\,\mathrm{Tr}(hM_g)$	–	8	16	32	64

Figure 3: Four calibrated bridges. Each M_g has four equal-mass magic corners; its selected e_g is the next flag.

Every transported label. Encode a unitary directly, a projection P by $2P - I$, a positive contraction T_+ by a unitary u_+ with $T_+ = (2I + u_+ + u_+^*)/4$, and a general contraction by an average of two unitaries. Let Λ be the resulting finite unitary label list that must be transported. For every $k \in \Lambda$, $r = 0, \ldots, 4$, introduce a unitary $u_{r,k}$ and impose

$$(u_{r,k} - I)(I - e_r) = 0. \tag{16}$$

The unitary reduces e_r and is identity on its complement. For a typed label L, its corner copy L_r is explicitly decoded as

type of L	L_r
unitary	$e_r u_{r,L}$
projection	$e_r(I + u_{r,L})/2, \quad u_{r,L}^2 = I$
positive contraction	$e_r(2I + u_{r,L} + u_{r,L}^*)/4$
general contraction	$e_r(u_{r,L,+} + u_{r,L,-})/2$

Thus $(P_\circ)_4$ denotes the decoded root projection at e_4. The two encodings of a general contraction are transported separately when that label occurs at a left endpoint.

For each $g = 1, \ldots, 4$ and every $k \in \Lambda$ introduce a unitary $v_{g,k}$ with

$$v_{g,k} = I - (e_{g-1} + M_g) + e_{g-1}u_{g-1,k} + \sum_{a,b} \mathfrak{x}_g^a \mathfrak{z}_g^b e_g u_{g,k} \mathfrak{z}_g^{-b} \mathfrak{x}_g^{-a}. \tag{17}$$

For this bridge, apply Lemma 4.2 to the six projections

$$F_1, F_2, F_3, G_{12}(1), G_{13}(1), G_{23}(1)$$

and the $|\Lambda|$ graph projections $G_{12}(v_{g,k})$. Thus there are $6 + |\Lambda|$ refinements of each retained sign question per bridge, on each party. Each label has its own duplicated question; the unchanged opposite questions are always retained.

The six frame commutators make $w_g = I_3 \otimes \widetilde{w}_g$. The commutator of the original sign with $G_{12}(v_{g,k})$, in its (c_0, f) block, is precisely

$$w_g u_{g-1,k} = u_{g,k} w_g. \tag{18}$$

For clarity the sign has off-diagonal block w_g between flag and selected magic direction, while the block unitary has the respective diagonal blocks $u_{g-1,k}, u_{g,k}$. Their commutator equates these two products. Consequently $L_4 v = v L_0$ for every decoded transported label, and for every expression built from them. Here L_r means its decoded copy at e_r, whose corner identity is e_r. The same actual w_g works for every label.

In detail. Equation (16) says $u_{r,k}(I - e_r) = I - e_r$. Multiplying by $u_{r,k}^*$ gives $u_{r,k}^*(I - e_r) = I - e_r$, and the adjoint of this is $(I - e_r)u_{r,k} = I - e_r$. Hence $u_{r,k}$ commutes with e_r and is the identity on $(I - e_r)P_A\mathcal{H}_A$. On the active space of bridge g, the (a, b) term of (17) is $t_{ab}(e_g u_{g,k} e_g)t_{ab}^*$, because $\mathfrak{x}_g, \mathfrak{z}_g$ are reflections and $\mathfrak{x}_g^a \mathfrak{z}_g^b e_g = t_{ab}$. So $v_{g,k}$ is block diagonal for

$e_{g-1} \oplus \bigoplus_{ab} m_{g,ab}$, with the block $u_{g-1,k}$ on the flag and transported copies of $u_{g,k}$ on the magic corners; in particular it reduces e_{g-1} and e_g. The six frame refinements make the sign $\mathcal{X}_0$ of bridge g commute with F_1, F_2, F_3 and $G_{ij}(1)$. These generate $M_3 \otimes I$ in the frame coordinates of Section 4.2, whose commutant is $I_3 \otimes B(\mathcal{H}_{\mathrm{base}})$, and the corners are frame-bound; hence $w_g = I_3 \otimes \widetilde{w}_g$. The refinement by $G_{12}(v_{g,k}) = \frac{1}{2}(F_1 + F_2 + e_{12} \otimes v_{g,k} + e_{21} \otimes v_{g,k}^*)$ makes $\mathcal{X}_0$ commute with it, and the $(1,2)$ frame block gives $[\widetilde{\mathcal{X}}_0, v_{g,k}] = 0$ on the base. With respect to $e_{g-1} \oplus e_g$,

$$\widetilde{\mathcal{X}}_0 = \begin{pmatrix} 0 & \beta^* \\ \beta & 0 \end{pmatrix}, \qquad v_{g,k} = \begin{pmatrix} u_{g-1,k} & 0 \\ 0 & u_{g,k} \end{pmatrix},$$

and the lower-left block of the commutator is $\beta u_{g-1,k} - u_{g,k}\beta$. This is (18). Induction over g gives $u_{4,k}v = vu_{0,k}$. Multiplying on the left by $u_{4,k}^*$ and on the right by $u_{0,k}^*$ gives $u_{4,k}^* v = v u_{0,k}^*$. The decoded copies in the table are polynomials in $e_r, u_{r,L}, u_{r,L}^*$ with the same coefficients for every r, and $e_4 v = v = v e_0$. Hence $L_4 v = v L_0$ and $L_4^* v = v L_0^*$, and the same holds for sums and products of decoded labels.

In the application the transported typed list is exactly

$$P_\circ,\ U_x,\ U_y,\ P_x,\ P_y,\ W,\ C_{\mathrm{ctr}},\ Z_{\mathrm{clk}},\ J_{\mathrm{W}},\ K_{\mathrm{W}}. \tag{19}$$

Table 4 records the label types. Here $P_\circ, P_x, P_y$ are projections, C_{ctr} is positive, and the other six are unitary. Thus one unitary encoding per typed label suffices, so $|\Lambda| = 10$. There are fifty support equations, forty block equations and sixteen refinements per bridge per party. Source-only labels $P_{\mathrm{ret}}, T_{\mathrm{inv}}, P_c$ do not occur at left endpoints of an A_α relation and need no transport.

Label	Type	Transported?	Use
$P_\circ$	projection	yes	Fock root
U_x, U_y	unitary	yes	coordinate increments
P_x, P_y	projection	yes	boundaries
W	unitary	yes	clock
C_{ctr}	positive contraction	yes	counter
$Z_{\mathrm{clk}}, J_{\mathrm{W}}, K_{\mathrm{W}}$	unitary	yes	Weyl clock
P_{ret}	projection	no	pulse
T_{inv}	general contraction	no	reader inverse
P_c	projection	no	tableau
a_α	general contraction	scaffold only	lifted creation

Table 4: Typed application labels. Transported labels are copied through all bridges for use at a left endpoint.

Extraction and exact translation. Introduce four centralizer contractions $a_\alpha = (a_{\alpha,+} + a_{\alpha,-})/2$ and impose $a_\alpha = e_4 a_\alpha e_0$. On the root corner extract $A_\alpha = v^* a_\alpha$. The adjoint of (14) gives $hv^* = v^* h/16$, so $h_0 A_\alpha = A_\alpha h_0/16$, where $h_0 = 121 e_0 h e_0$. Supply the following polynomials to the graph compiler:

Root identity	Centralizer identity
$A_\alpha^* A_\beta = \delta_{\alpha\beta} I$	$a_\alpha^* a_\beta = \delta_{\alpha\beta} e_0$
$\sum_\alpha A_\alpha A_\alpha^* = I - P_\circ$	$\sum_\alpha a_\alpha a_\alpha^* = e_4 - (P_\circ)_4$
$L A_\alpha = A_\beta B$	$L_4 a_\alpha = a_\beta B_0$
$L A_\alpha = 0$	$L_4 a_\alpha = 0$
Root-only polynomial	Same polynomial in source copies

Both implications hold. The range condition and $vv^* = e_4$ identify $A_\alpha^* A_\beta$ with $a_\alpha^* a_\beta$. Conjugating the range-sum equation by v uses the transported $P_\circ$. Transport of decoded adjoints also gives $v^* L_4 = L_0 v^*$. For an endpoint relation multiply by v and use $L_4 v = v L_0$; conversely use injectivity of v^* on e_4. No polynomial submitted to graph compilation contains v.

Expand the finite words, bind their parameters, apply graphs and triangles, and then Lemma 4.3. Retain all FMS and refinement questions. Pad the two question sets if necessary and sample uniformly from all pairs, accepting every unspecified pair. The result has rational question distribution and a Boolean predicate. The support arguments prove soundness on a nonzero corner, using Alice alone; mirror equations are unnecessary.

In detail. Put $\mathcal{H}_0 = e_0 P_A \mathcal{H}_A$ and let h_0 be the restriction of $121 e_0 h_A e_0$ to $\mathcal{H}_0$. It is positive and trace class, it is faithful because h_A is faithful on $P_A \mathcal{H}_A$ and commutes with e_0, and it has trace one by (15); in particular $\mathcal{H}_0 \neq 0$. Every source copy L_0 maps $\mathcal{H}_0$ into itself and commutes with h_0, because $u_{0,L}$ commutes with e_0 and with h_A. The declared types hold: $e_0 u_{0,L}$ is unitary on $\mathcal{H}_0$; $e_0(I + u_{0,L})/2$ is a projection when $u_{0,L}^2 = I$; $e_0(2I + u + u^*)/4$ is a positive contraction; and an average of two unitaries is a contraction. The contraction $a_\alpha = e_4 a_\alpha e_0$ commutes with h_A, so $A_\alpha = v^* a_\alpha$ maps $\mathcal{H}_0$ into itself, and

$$h_0 A_\alpha = 121\, e_0 h_A v^* a_\alpha = \tfrac{121}{16}\, v^* h_A a_\alpha = \tfrac{121}{16}\, v^* a_\alpha e_0 h_A e_0 = A_\alpha h_0/16,$$

using $e_0 v^* = v^*$, $h_A v^* = v^* h_A/16$, $a_\alpha = a_\alpha e_0$ and $[e_0, h_A] = 0$. The translations are exact. For the creation equations, $A_\alpha^* A_\beta = a_\alpha^* v v^* a_\beta = a_\alpha^* e_4 a_\beta = a_\alpha^* a_\beta$. For the range sum, $\sum_\alpha A_\alpha A_\alpha^* = v^*(\sum_\alpha a_\alpha a_\alpha^*)v$. If $\sum_\alpha a_\alpha a_\alpha^* = e_4 - (P_\circ)_4$, this equals $e_0 - v^*(P_\circ)_4 v = e_0 - (P_\circ)_0$; conversely, conjugation by v recovers the centralizer form, because $vv^* = e_4$ and $a_\alpha = e_4 a_\alpha$. For an endpoint equation, $v^* L_4 = L_0 v^*$ gives $L_0 A_\alpha - A_\beta B_0 = v^*(L_4 a_\alpha - a_\beta B_0)$. The bracket has range in $e_4 P_A \mathcal{H}_A$, where v^* is isometric, so either side vanishes exactly when the other does. Sums $\sum_\nu B_\nu A_{\alpha_\nu} C_\nu$ translate in the same way, with $(B_\nu)_4$ on the left. Root-only polynomials involve only source copies. Every application equation therefore holds as an operator identity on $\mathcal{H}_0$.

One simultaneous realization. Given a root realization with density h_0, take

$$h_{\mathrm{sc}} = \frac{1}{121}\,\mathrm{diag}(h_0;\, 2h_0^{\oplus 4};\, 4h_0^{\oplus 4};\, 8h_0^{\oplus 4};\, 16h_0^{\oplus 4}). \tag{20}$$

Use identity corner identifications in the bridges and $a_\alpha = |e_4\rangle\langle e_0| \otimes A_\alpha$. Its commutation with h_{sc} is exactly the modular relation. Every positive eigenspace of a faithful trace-class density is finite-dimensional. In each such block the polar partial isometry of a contraction T extends to a unitary V_{pol}, and

$$T = (U_+ + U_-)/2, \qquad U_\pm = V_{\mathrm{pol}}(|T| \pm i\sqrt{I - |T|^2}).$$

Direct sums give centralizer encodings for all four a_α at once. The root isometries A_α are never themselves written as averages of centralizer unitaries. The contraction completion takes place on the scaffold. Encode each source contraction inside the source density blocks, choose $2P - I$ for projections, and use

$$u_+ = 2T_+ - I + 2i\sqrt{T_+(I - T_+)}$$

for a positive contraction T_+. Copy these same choices through the four corner identifications and extend by identity outside the corners. The block equations now hold simultaneously. Append the scalar qutrit frame, use the graph formulas, and take actual joint products for all refinements. Their block unitaries commute with the coarse FMS effects. Separate labels

need not commute. Finally append the finite maximally entangled row factor. All tests hold with one state and one assignment. The scaffold and ancillary dimensions are finite, so they add only finite entropy.

In detail. Let $(\mathcal{H}_0, h_0)$ be a normal realization, and put $\mathcal{H}_{\mathrm{sc}} = \mathbb{C}^{17} \otimes \mathcal{H}_0$ with the density (20). It is faithful and trace class, of trace $(1 + 8 + 16 + 32 + 64)/121 = 1$. Its positive eigenspaces are spanned by corner vectors tensored with positive eigenvectors of h_0, so they are finite-dimensional, and a bounded operator commuting with h_{sc} maps each of them into itself by (N4d). On such a block $\mathcal{E}$ the contraction $T = a_\alpha|_{\mathcal{E}}$ has a polar decomposition $T = V|T|$. Because $\dim\ker T = \dim\ker T^*$ in finite dimension, V extends to a unitary V_{pol} with $T = V_{\mathrm{pol}}|T|$, and $|T| \pm i(I - |T|^2)^{1/2}$ are commuting unitaries. The direct sums of the resulting unitaries over all blocks are unitaries commuting with h_{sc}. Finite-dimensional blocks are needed here: a nonunitary isometry, such as a root isometry A_α, is an extreme point of the unit ball and is therefore not an average of two unitaries. That is why A_α is never encoded directly. The other encodings are functions of operators commuting with h_0, so they commute with h_0. The copies $u_{r,k} = |e_r\rangle\langle e_r| \otimes U_k + (I - |e_r\rangle\langle e_r|) \otimes I$ satisfy (16); the block unitaries (17) act as U_k on every corner of bridge g and as I elsewhere; and $w_g = |e_g\rangle\langle e_{g-1}| \otimes I$ gives (18). The translated equations hold because, for example, $L_4 a_\alpha - a_\beta B_0 = |e_4\rangle\langle e_0| \otimes (LA_\alpha - A_\beta B)$.

The strategy is explicit. Take $\mathcal{H}_A = \mathcal{H}_B = \mathbb{C}^3 \otimes \mathcal{H}_{\mathrm{sc}} \otimes \mathbb{C}^{d_{\max}}$, the density $h = (I_3/3) \otimes h_{\mathrm{sc}} \otimes (I_{d_{\max}}/d_{\max})$ and the pure state Ω_h of (N8), formed in a product eigenbasis. Every synchronized question of Alice measures a projection commuting with h: corner, frame, graph, triangle and binding projections built from the centralizer unitaries by the formulas of Section 4.2; the FMS cell, activity and corner projections; and the row projections of Lemma 4.3. Bob measures their transposes. By (N8) every matching test passes. Every forbidden pair between centralizer events has probability $\mathrm{Tr}(hQR) = 0$, because the defining identities give $QR = 0$. The FMS sign questions are not in the centralizer. For bridge g, Alice's sign observable for T_x is the reflection $|f\rangle\langle c_x| + |c_x\rangle\langle f|$ on the span of the flag corner e_{g-1} and the corner c_x, tensored with the identity of $\mathcal{H}_0$. Bob's sign observable for B_y is $|f\rangle\langle r_y| + |r_y\rangle\langle f|$, with $r_y = (c_0 + (-1)^y c_1)/\sqrt{2}$. Their retained answers are the remaining column-zero and row-zero projections. If μ is the mass of e_{g-1}, each magic corner has mass 2μ, and $\langle c_x, r_y\rangle = \sigma_{xy}/\sqrt{2}$. So, up to the frame and row factors, the compressed state is $\Omega_{xy} = \mu^{1/2}(|f, f\rangle + \sigma_{xy}|c_x, r_y\rangle) \otimes \Omega_{h_0}$. Substitution gives $(\mathcal{X}_x \otimes I)\Omega_{xy} = \sigma_{xy}(I \otimes \mathcal{Y}_y)\Omega_{xy}$, so every forbidden sign pair has probability zero; the calibrated rules against C_0 and R_0 hold by construction. The FMS effects are corner operators tensored with the identity of $\mathcal{H}_0$. They commute with every block unitary $v_{g,k}$ and with the frame, on Alice's side and, after transposition in the product eigenbasis, on Bob's. The refinements $E_a P^{\mathrm{lab}}$ of Lemma 4.2 are therefore POVMs. The marginal entropy is $\log_2 3 + S(h_{\mathrm{sc}}) + \log_2 d_{\max}$, where $S(h_{\mathrm{sc}}) = H(\mu_{\mathrm{sc}}) + S(h_0)$ for the vector μ_{sc} of scaffold masses. □

4.5 A dimension-independent completeness estimate

The exact construction gives a useful quantitative statement. Keep the label types, centralizer conditions, contractivity and modular covariance exact, but allow residuals r_j in the application equations of Section 4.4. Write $\|T\|_{2,h}^2 = \mathrm{Tr}(hT^*T)$. The constructed strategy satisfies

$$1 - \omega(G, \mathcal{S}) \le C_{\mathrm{comp}} \sum_j \|r_j\|_{2,h}^2, \tag{21}$$

where C_{comp} depends only on the finite presentation and the compiler.

Here is the proof, including the density factors. The same scaffold, exact unitary encodings and actual products make all structural tests perfect. Only the final application row-zero tests may lose. Let p_j be the lifted polynomial residual. For a source or root-polynomial row it is

the source copy of r_j; for an endpoint row it is vr_j; for a range row it is vr_jv^*. Consequently

$$\|p_j\|_{2,h_{\mathrm{sc}}}^2 = \frac{\beta_j}{121}\|r_j\|_{2,h}^2, \qquad \beta_j = \begin{cases} 16, & \text{range row at } e_4, \\ 1, & \text{otherwise.} \end{cases}$$

For the endpoint case use $v^*v = e_0$. For the range case use $v^*h_{\mathrm{sc}}v = 16h/121$. These are equalities with no density-gap estimate. All traces here are finite, because h_{sc} is trace class and every p_j is bounded. The identity $\mathrm{Tr}(h_{\mathrm{sc}}p_jp_j^*) = \mathrm{Tr}(h_{\mathrm{sc}}p_j^*p_j)$ used next follows from $[p_j, h_{\mathrm{sc}}] = 0$ and cyclicity of the trace.

The qutrit graph row S_j has off-diagonal blocks p_j, p_j^*. Since p_j commutes with h_{sc},

$$\mathrm{Tr}(h_{\mathrm{old}}S_j^2) = \frac{1}{3}\,\mathrm{Tr}[h_{\mathrm{sc}}(p_j^*p_j + p_jp_j^*)] = \frac{2\beta_j}{363}\|r_j\|_{2,h}^2.$$

Apply (13). For uniform Q-by-Q sampling one may take

$$C_{\mathrm{comp}} = \max_j \frac{8L_j^2\beta_j}{363d_j^3Q^2}. \tag{22}$$

Summing event probabilities gives an upper bound even if rejecting rules overlap. The dimension, clock order, density spectrum and spectator dimensions do not enter this constant. No continuity of polar completions or approximate dense-range cancellation is used: the structural realization is built exactly. This is a completeness estimate for constructed strategies under the stated exact structural hypotheses. It is not a robust extraction claim for arbitrary nearly perfect strategies.

5 The rooted tree and computation squares

5.1 Normality removes every rootless sector

Require four isometries and a root projection with

$$A_{ab}^*A_{cd} = \delta_{ac}\delta_{bd}I, \qquad \sum_{ab} A_{ab}A_{ab}^* = I - P_\circ, \qquad h_0A_{ab} = A_{ab}h_0/16. \tag{23}$$

Here h_0 is a faithful density on the extracted root-corner Hilbert space. For $w = w_1 \cdots w_m$ set $A_w = A_{w_1} \cdots A_{w_m}$ (the empty product is I) and put $Q_m = \sum_{|w|=m} A_wA_w^*$. Orthogonality and trace cyclicity give $\mathrm{Tr}(h_0Q_m) = 4^{-m}$. The projections Q_m decrease. Their strong limit has zero h_0-weight, hence is zero by faithfulness. Also $Q_m - Q_{m+1} = \sum_{|w|=m} A_wP_\circ A_w^*$. Therefore

$$\mathcal{H} = \bigoplus_w A_w\mathcal{K} \simeq \ell^2(\mathcal{W}) \otimes \mathcal{K}, \qquad \mathcal{K} = P_\circ\mathcal{H} \neq 0. \tag{24}$$

Here $\mathcal{W}$ denotes finite words in the four-letter alphabet $\{0,1\}^2$. The modular equation makes h_0 commute with every $A_wP_\circ A_w^*$. Since $\mathrm{Tr}(h_0P_\circ) = 3/4$, transporting its root block gives

$$h_0 = h_F \otimes \sigma, \qquad h_F = \frac{3}{4}\sum_w 16^{-|w|}|w\rangle\langle w|, \qquad \mathrm{Tr}\,\sigma = 1. \tag{25}$$

In particular

$$S(h_F) = \log_2(4/3) + 4/3. \tag{26}$$

Indeed the depth-m mass is $(3/4)4^{-m}$ and its mean depth is $1/3$; the log probability at each word is $\log_2(4/3) + 4m$. This proves the formula and excludes a rootless normal summand.

5.2 Each depth is a finite square

For $w = (a_1, b_1) \cdots (a_m, b_m)$ define

$$x(w) = \sum_{j=1}^{m} a_j 2^{j-1}, \qquad y(w) = \sum_{j=1}^{m} b_j 2^{j-1}.$$

The first letter is the least significant bit. Prefixing A_{ab} maps (x, y) to $(2x + a, 2y + b)$. Depth m is a $2^m \times 2^m$ square. Require centralizer unitaries U_x, U_y and projections P_x, P_y satisfying, for $a, b \in \{0, 1\}$,

$$U_x A_{0b} = A_{1b}, \qquad U_x A_{1b} = A_{0b} U_x, \qquad U_x P_\circ = P_\circ, \tag{27}$$
$$U_y A_{a0} = A_{a1}, \qquad U_y A_{a1} = A_{a0} U_y, \qquad U_y P_\circ = P_\circ, \tag{28}$$
$$P_x A_{0b} = A_{0b} P_x, \qquad P_x A_{1b} = 0, \qquad P_x P_\circ = P_\circ, \tag{29}$$
$$P_y A_{a0} = A_{a0} P_y, \qquad P_y A_{a1} = 0, \qquad P_y P_\circ = P_\circ. \tag{30}$$

Induction on word length identifies U_x and U_y as cyclic increments of x and y respectively, acting identically on $\mathcal{K}$. The same induction makes P_x, P_y the indicators of $x = 0, y = 0$. In particular U_x, U_y commute, and

$$P_{x,\text{last}} = U_x^* P_x U_x, \qquad P_{y,\text{last}} = U_y^* P_y U_y$$

mark the final column and row. These are conclusions on the full root multiplicity, not just selected vectors.

5.3 A finite deterministic tableau

Choose a deterministic radius-one cellular automaton on the half-line. Its finite alphabet Γ includes start s, blank b and halt H, and its update is $\Phi(\ell, c, r, \varepsilon)$ with left boundary symbol $\#$ and external pulse $\varepsilon \in \{0, 1\}$. The initial row is $s, b, b, \ldots$. We use these conventions:

- $\Phi(b, b, b, \varepsilon) = b$ for both pulse values.
- The initial start state ignores the pulse at time zero. Later waiting-front states respond to positive-time pulses.
- Cells passed by the waiting front keep their unary marks. Frozen cells and all states of the later input-checking computation ignore subsequent pulses.
- A transition using the pulse at time y produces row $y + 1$. Thus the pulse at time q records the front at coordinate q in that next row.

Here Φ names the update function; f in the following table is a front-state symbol. Here is an explicit front layer, with alphabet $\{s, b, f, z, H\}$. The right neighbour is unused in these six rules:

condition	next centre
$c = s$	b
$c = f$	z if $\varepsilon = 1$, b if $\varepsilon = 0$
$c = b,\ \ell = s$	f
$c = b,\ \ell = f$	f if $\varepsilon = 0$, b if $\varepsilon = 1$
$c = z$	z
$c = H$	H

All remaining blank centres stay blank. At time $y \geq 1$, until a return, the unique front is at $x = y$. The first positive pulse at q writes z at $(q, q + 1)$ and suppresses the front's successor. The frozen mark's distance from the left boundary is the unary input q; blank cells between the boundary and the mark represent its unary positions. Pulse-dependence is confined to neighbourhoods containing the front. In particular a blank right of f may read the pulse.

Add a persistent unary bit $u \in \{0,1\}$, initially zero, with $u' = 1$ when the old front centre is s or f, and $u' = u$ otherwise. Thus passed cells retain their unary marks on a separate track, even though their front-layer component is blank. Take the product alphabet $\{s, b, f, z, H\} \times \{0,1\} \times \mathcal{T}$, with a finite TM alphabet $\mathcal{T} = \mathcal{A} \times (Q \cup \{\text{no head}\})$ for the fixed one-tape half-line program in Section 7. All TM cells start blank without a head. The transition $f \to z$ initializes its head in the start state at that cell. Thereafter a centre carrying (a, p) writes the symbol prescribed by $\delta(p, a, u)$ and retains a head only for a stationary move. A head moving right (left) is received by the right (left) neighbour, whose old tape symbol is retained. A centre receiving no head retains its symbol without a head. Invalid multiple incoming heads may be sent to a fixed error state; they never occur from this initialization. The left boundary # is read as a tape end marker and no transition moves left of it. The frozen front track is read-only input for this TM. A halting head writes a designated halt symbol H in the combined alphabet and keeps it. These prescriptions give a finite radius-one lookup table effectively from the finite TM transition table; all subsequent TM transitions ignore the pulse. Denote this finite product alphabet, with the halt convention, by Γ. In the combined alphabet, s and b abbreviate the start and fully blank product states. Collapse any product state with a halting head to the single absorbing symbol H.

Blank quiescence holds before a front or head arrives. The s rule ignores the time-zero pulse and seeds f at time one via its right neighbour. The mark is permanent and later pulses cannot restart a head. Finally, all disturbances start at the origin, or at the marked front within its radius-one cone, so no nonblank cell can occur at $x > y$. These checks give all four conventions and the blank right ghost used below.

Let $(P_c)_{c\in\Gamma}$ be a centralizer PVM and P_{ret} a centralizer projection, with $P_{\text{ret},0} = I - P_{\text{ret}}$, $P_{\text{ret},1} = P_{\text{ret}}$. Require

$$P_s P_x P_y = P_x P_y, \qquad P_b(I - P_x)P_y = (I - P_x)P_y, \qquad P_H = 0. \tag{31}$$

Define centralizer polynomials

$$L_{\#} = P_x, \qquad L_c = U_x P_c U_x^*(I - P_x),$$
$$N_r = U_x^* P_r U_x (I - P_{x,\text{last}}) + \mathbf{1}_{\{r=b\}} P_{x,\text{last}}.$$

For every $a \in \Gamma$ impose

$$\left[U_y^* P_a U_y - \sum_{\Phi(\ell,c,r,\varepsilon)=a} L_\ell P_c N_r P_{\text{ret},\varepsilon} \right] (I - P_{y,\text{last}}) = 0. \tag{32}$$

The factors have the displayed order. They need not commute globally. The factor $I - P_{y,\text{last}}$ removes a spurious transition from the final row to row zero. The ghost to the right of the last column is blank.

Lemma 5.1 (Definite computation)**.** *Suppose a root vector ξ belongs to a sector where the clock pulse on every coordinate has a specified scalar value. Then every P_c acts on $A_w\xi$ as the indicator of the genuine classical computation at $(x(w), y(w))$.*

Proof. Work at a fixed depth m and induct on time, for every spatial coordinate at once. Equation (31) gives row zero. On an induction vector every factor $L_\ell, P_c, N_r, P_{\text{ret},\varepsilon}$ acts by its classical zero or one. Their ordered product therefore selects exactly one update. Equation (32) gives $U_y^* P_a U_y$ with the correct scalar on that vector. Multiplying by U_y gives the next-row assertion.

The half-line and the square agree throughout this induction. At time y no disturbance has travelled beyond coordinate y. In particular the ghost cell 2^m is blank for all times $y \le 2^m - 1$,

by radius one and quiescence under either pulse. The left ghost is always #. This proves the genuine-prefix assertion. A halting event at finite coordinates and time occurs in every sufficiently large square and contradicts $P_H = 0$. □

The tableaux at distinct depths need not intertwine with the creation operators. Determinism forces each of them to the same finite prefix; soundness needs only one sufficiently large square. Every depth has positive normal weight by (25).

5.4 A worked front and mask example

Table 5 is rendered by `review/generate_toy.py` from `tableau(3,3)[3]`. Its d denotes a passed cell and m a frozen mark. These persistent annotations differ from the minimal front layer above, whose passed cells are blank and whose mark is z; both put the mark at $(3, 4)$. The pulse at time zero is ignored by s. The pulse at $y = 3$ produces the mark in row 4, and the later pulse at $y = 6$ has no effect.

y	pulse	0	1	2	3	4	5	6	7	update mask
0	1	s	b	b	b	b	b	b	b	1
1	0	d	f	b	b	b	b	b	b	1
2	0	d	d	f	b	b	b	b	b	1
3	1	d	d	d	f	b	b	b	b	1
4	0	d	d	d	m	b	b	b	b	1
5	0	d	d	d	m	b	b	b	b	1
6	1	d	d	d	m	b	b	b	b	1
7	0	d	d	d	m	b	b	b	b	0

Table 5: Depth $m = 3$, period $q = 3$: the decoder's 8×8 toy tableau. P_x marks column 0, $P_{x,\mathrm{last}}$ column 7, P_y row 0 and $P_{y,\mathrm{last}}$ row 7. The last-column neighbour is replaced by blank; the last-row mask suppresses the toroidal $7 \to 0$ update.

6 A finite-order clock and its period reader

6.1 A determinant supplies a finite witness

Add centralizer unitaries $J_{\mathrm{W}}, K_{\mathrm{W}}, Z_{\mathrm{clk}}$ with

$$\begin{gathered} J_{\mathrm{W}} K_{\mathrm{W}} = Z_{\mathrm{clk}} K_{\mathrm{W}} J_{\mathrm{W}}, \qquad [Z_{\mathrm{clk}}, J_{\mathrm{W}}] = [Z_{\mathrm{clk}}, K_{\mathrm{W}}] = 0, \\ [J_{\mathrm{W}}, A_{ab}] = [K_{\mathrm{W}}, A_{ab}] = [Z_{\mathrm{clk}}, A_{ab}] = 0. \end{gathered} \tag{33}$$

A unitary commuting with an isometry also commutes with its adjoint: its inverse commutes with the isometry, and taking adjoints gives the assertion. Thus these labels reduce $\mathcal{K} = P_\circ \mathcal{H}$ and commute with σ. Their action on every word sector is their root action.

Every positive eigenspace of σ is finite-dimensional. It reduces $J_{\mathrm{W}}, K_{\mathrm{W}}, Z_{\mathrm{clk}}$ and decomposes into Z_{clk} eigenspaces. On a nonzero such eigenspace of dimension d, with phase ζ, the Weyl relation gives

$$\det(J_{\mathrm{W}} K_{\mathrm{W}}) = \zeta^d \det(K_{\mathrm{W}} J_{\mathrm{W}}), \qquad \zeta^d = 1. \tag{34}$$

Since the positive eigenspaces exhaust $\mathcal{K}$, Z_{clk} has pure point spectrum there and every phase is a root of unity. At least one sector is nonzero. Conversely a q-dimensional Weyl pair realizes every primitive qth root, with density I_q/q.

6.2 The clock and a counter without a positive lower bound

Take a centralizer unitary W and a positive centralizer contraction C_{ctr} with

$$WP_\circ = P_\circ, \qquad WA_{ab} = A_{ab}Z_{\mathrm{clk}}^b W^2, \tag{35}$$

$$C_{\mathrm{ctr}}P_\circ = P_\circ, \qquad C_{\mathrm{ctr}}A_{ab} = 2^{-b}A_{ab}C_{\mathrm{ctr}}^2. \tag{36}$$

Also require Z_{clk} to commute with $U_x, U_y, P_x, P_y, W, C_{\mathrm{ctr}}, P_{\mathrm{ret}}, T_{\mathrm{inv}}$ and all P_c. Induction on word length, using the coordinate recurrence, gives

$$W|_{(x,y)} = Z_{\mathrm{clk}}^y, \qquad C_{\mathrm{ctr}}|_{(x,y)} = 2^{-y}I. \tag{37}$$

In particular C_{ctr} is injective even though its spectrum accumulates at zero.

For a centralizer projection P_{ret} and contraction T_{inv}, impose

$$(W-I)P_{\mathrm{ret}} = 0, \qquad T_{\mathrm{inv}}(W-I) = \tfrac{1}{2}C_{\mathrm{ctr}}(I-P_{\mathrm{ret}}). \tag{38}$$

The first equation puts $\operatorname{ran} P_{\mathrm{ret}}$ in $\ker(W-I)$. For ξ in that kernel the second gives $C_{\mathrm{ctr}}(I - P_{\mathrm{ret}})\xi = 0$. Injectivity of C_{ctr} gives $P_{\mathrm{ret}}\xi = \xi$. Thus

$$P_{\mathrm{ret}} = \operatorname{proj}\ker(W-I). \tag{39}$$

On a root phase sector of order q, the pulse is exactly $\mathbf{1}_{\{q|y\}}$ at every spatial coordinate. It cannot vanish spuriously or postpone the first return.

6.3 A bounded inverse on every prime clock

Lemma 6.1 (A good primitive phase)**.** *For every prime q there is a primitive qth root ζ such that*

$$|\zeta^t - 1| \geq 2^{-t-1} \quad (t \geq 1,\ q \nmid t).$$

Proof. For $1 \leq r < q$,

$$|e^{2\pi i r/q} - 1| \geq 4\min(r, q-r)/q.$$

Fix $1 \leq t < q$. Multiplication by t permutes nonzero residues. A phase $e^{2\pi ik/q}$ violating the required inequality must satisfy $\min(tk \bmod q, q - (tk \bmod q)) < q2^{-t-3}$. There are at most $q2^{-t-2}$ such residues. Summing over t excludes fewer than $q/4$ choices, while $q/4 < q-1$. One of the $q-1$ nontrivial phases works. For $t \geq q$ reduce to a nonzero residue $r < q$; its lower bound 2^{-r-1} is at least 2^{-t-1}. □

For this phase set

$$T_{\mathrm{inv}}|_{(x,y)} = \begin{cases} 2^{-y-1}(\zeta^y - 1)^{-1}I, & q \nmid y, \\ 0, & q \mid y. \end{cases} \tag{40}$$

Lemma 6.1 gives $\|T_{\mathrm{inv}}\| \leq 1$. Equations (37) and (39) then verify (38). The game coefficients remain rational: the phase belongs to the strategy, not the predicate.

7 The reduction

Fix any decidable predicate $R(e,n,t)$ and the set $S = \{e : \exists n\, \forall t\ R(e,n,t)\}$. We effectively choose the cellular automaton in Section 5 to run the following program, with e hardwired.

Read a positive-time return. At time y a waiting front is at position y. Ignore the time-zero pulse. At the first positive-time pulse, freeze the front and retain its coordinate as unary input q. With the transition convention above the mark is written into row $q+1$ at position q. Test q. If $q=1$ or q is composite, halt. Otherwise compute the unique $n\ge 0$ with $q=p_{n+1}$, the $(n+1)$st prime. Then evaluate $R(e,n,0), R(e,n,1),\dots$ and halt at the first false answer. All later computation states ignore the external pulse.

For a Σ_2^0-complete choice of S, there is no computable bound on a sufficient clock order as a function of the input game: such a bound would leave only finitely many possible n in this reduction, making $\exists n\,\forall t\ R(e,n,t)$ a Π_1^0 predicate. Indeed one tests, for every finite time cutoff, whether some one of those finitely many candidates survives all checks so far; if none survives forever, all have failed by a common finite cutoff. This contradicts Σ_2^0-hardness.

This finite program has an effective finite radius-one simulation. The initial-front rule, frozen unary track and ordinary tape/head tracks give the stated boundary conventions. We do not require a bound on its running time. Each completed false check causes the halt symbol after finitely many updates.

Include the Fock, coordinate, clock, reader and tableau equations above, all typing relations, and the stated Z_{clk} commutators. They form an admissible finite presentation. For example their transported polynomials include

$$W_4 a_{ab} = a_{ab} Z_{\mathrm{clk},0}^{b} W_0^2, \qquad C_{\mathrm{ctr},4} a_{ab} = 2^{-b} a_{ab} C_{\mathrm{ctr},0}^2, \qquad J_{\mathrm{W},4} a_{ab} = a_{ab} J_{\mathrm{W},0}.$$

The source-only return equations and tableau equations are centralizer polynomials. Lemma 4.4 therefore computes a finite rational Boolean game G_e.

Theorem 7.1. *The map $e\mapsto G_e$ satisfies*

$$G_e\in \mathrm{PERF}_{\mathrm{qs}} \iff G_e\in \mathrm{PERF}_{\mathrm{finS}} \iff \exists n\,\forall t\ R(e,n,t).$$

No G_e has a perfect finite-dimensional strategy.

Proof. A perfect normal strategy extracts a nonzero faithful root realization. The determinant argument supplies a nonzero root phase sector of finite order q. By (33) its word descendants persist at all depths. Equations (35)–(39) force the exact return sequence, with its first positive return at q. Lemma 5.1 needs only that the initial root vector lie in this Z_{clk}-eigenspace: the clock pulse is then scalar on all its word descendants. The imposed $[Z_{\mathrm{clk}},P_c]=0$ relations are redundant for this induction. An invalid q halts. For valid $q=p_{n+1}$ any false $R(e,n,t)$ also halts. A sufficiently large square would contain that halt, contrary to $P_H=0$. Hence the extracted n witnesses membership in S.

Conversely take a witness n and set $q=p_{n+1}$. Use (25) with $\sigma=I_q/q$, a good primitive phase and its Weyl pair, the canonical coordinate permutations, (37), (39), (40), and the classical nonhalting tableau. Every required operator is bounded and commutes with the density when so specified. This is a simultaneous realization. On $\mathbb{C}^q$ one may take $J_{\mathrm{W}}|j\rangle = \zeta^j|j\rangle$, $K_{\mathrm{W}}|j\rangle = |j+1 \bmod q\rangle$, and $Z_{\mathrm{clk}}=\zeta I$. Its root entropy and largest Schmidt probability are

$$S(h)=\log_2 q+\log_2(4/3)+4/3, \qquad p_{\max}=3/(4q). \tag{41}$$

The compiler adds finite factors, giving a perfect finite-entropy strategy.

The tensor-product witness can be written without an implicit extension. Let $\mathcal{H}_F=\ell^2(\mathcal{W})$ and let $\mathcal{H}_{\mathrm{sc}}=\mathbb{C}^{17}$ index the scaffold corners. Take

$$\mathcal{H}_A=\mathcal{H}_B=\mathcal{H}_{\mathrm{sc}}\otimes\mathcal{H}_F\otimes\mathbb{C}^q\otimes\mathbb{C}^3\otimes\mathbb{C}^{d_{\max}}.$$

If $\mu=(1,2,2,2,2,4,4,4,4,8,8,8,8,16,16,16,16)/121$, use the pure vector

$$|\psi\rangle = |\sqrt{\mu}\rangle\!\rangle\otimes|\sqrt{h_F}\rangle\!\rangle\otimes|\Phi_q\rangle\otimes|\Phi_3\rangle\otimes|\Phi_{d_{\max}}\rangle, \quad |\sqrt{\mu}\rangle\!\rangle=\sum_i \sqrt{\mu_i}|i,i\rangle.$$

The Fock factor is $\sum_w \sqrt{(3/4)16^{-|w|}}|w,w\rangle$. Alice uses the scaffold FMS effects, actual graph projections, refinements and column partitions of Lemma 4.4; Bob uses the prescribed opposite FMS effects and the transposed transferred centralizer products. The row factor is the finite maximally entangled ancilla of Lemma 4.3. These simultaneous effects were checked in the compiler proof. The marginal entropy is $H(\mu) + S(h_F) + \log_2 q + \log_2 3 + \log_2 d_{\max} < \infty$.

Finally finite-dimensional perfect play would extract four isometries with mutually orthogonal ranges on one nonzero finite-dimensional root space. Their range dimensions would sum to four times the dimension of that space, which is impossible. The argument applies to every input, whether or not it has a witness. □

Together with Section 3, this proves both completeness statements in Theorem A. The value-one promise is established next. For the promise version, the reduction maps every input into that promise; the unrestricted Σ_2^0 membership predicate restricts to it, giving promise-problem completeness. The construction can be made syntactically nonsynchronous and non-imitation on every input by adding a fresh binary question to each side, accepting every pair involving it. Either answer then has the same nonempty allowed-answer support against every opposite question, which violates the imitation separation condition. This addition does not change perfection or the value-one property.

Size of the finite game. Here is a coarse order-of-magnitude bound, independent of the witness q. Let $g = |\Gamma|$, let N_e count the named graph projections, rows and word prefixes after endpoint translation, and let b_e bound the bit length of their rational coefficients. The lookup table has at most $2(g+1)g^2$ entries, so unshared explicit polynomials give $N_e = \mathrm{poly}(g)$ for this fixed list of operator relations. Denominator clearing and signed repetitions give row dimension $d_{\max} \le 2^{O(N_e(b_e+\log(N_e+1)))}$. The scalar contexts, column questions and marginal bindings are polynomial in $N_e, d_{\max}$, so the padded question count Q_e and maximum answer count obey the same coarse exponential bound. An explicit predicate table is polynomial in those two counts and has the same exponential order. Thus the effective compilation depends on the hardwired program, never on its chosen prime or running time. For scale, the bundled minimal one-isometry example already has 748 rows and 27 991 questions per party; it is not a size claim for the full reduction. That example is the minimal compiler instance: four bridges, one modular isometry and no transported labels. It is the only compiled game. The full reduction game G_e, with the ten transported labels of (19) and the tableau equations, was not compiled.

8 Every constructed game has value one

The finite-dimensional approximations do not need a witness for e. Truncate the Fock tree to depths $0, \dots, m$ and choose a prime $q > 2^m$ with a good phase (Lemma 6.1). Use the normalized density

$$h^{[m]} = \frac{\sum_{|w|\le m} 16^{-|w|}|w\rangle\langle w|}{\sum_{j=0}^{m} 4^{-j}} \otimes \frac{I_q}{q}.$$

Every represented time is less than q. The decoder therefore remains in the waiting stage in every square and never halts. Let P_m denote the top-depth projection. Its mass is

$$t_m = \mathrm{Tr}(h^{[m]}P_m) = \frac{3}{4^{m+1}-1}. \tag{42}$$

Every endpoint identity $LA_\alpha = A_\beta B$ remains exact. Below top depth it is the canonical identity; on top depth both sides vanish because the endpoint expression B preserves depth.

The modular equations, the range-sum identity and all off-diagonal $A_\alpha^* A_\beta = 0$ also remain exact. The only failed original relations are the four diagonal isometry equations:

$$A_\alpha^* A_\alpha - I = -P_m \quad (\alpha = 00, 01, 10, 11). \tag{43}$$

All graph and triangle projections are defined from the actual products. Every structural graph, synchrony, binding, FMS, refinement and column test stays exact.

Here is the full row accounting. For each α let L_α be the denominator-clearing integer and d_α the column dimension of the row obtained by translating $a_\alpha^* a_\alpha - e_0 = 0$ into graph projections. These depend on the finite input presentation, not on m or q. In the scalar qutrit frame its defect S_α has off-diagonal blocks $-(P_m)_0$ and adjoint and zero diagonal blocks. The frame density is $I_3/3$ and the source scaffold mass is $1/121$, so

$$\mathrm{Tr}(h_{\mathrm{old}} S_\alpha^2) = \frac{2}{3}\frac{1}{121} t_m. \tag{44}$$

The same calculation applies separately to each of the four rows. Their row-zero events have distinct column questions and occur in no other failed rule. If the game has Q_e equally padded questions per party, (13) gives

$$1 - \omega(G_e, \mathcal{S}_m) \le C_e \frac{3}{4^{m+1} - 1}, \tag{45}$$

$$C_e = \frac{1}{Q_e^2}\frac{2}{363} \sum_{\alpha \in \{00,01,10,11\}} \frac{4L_\alpha^2}{d_\alpha^3}. \tag{46}$$

If the four row constructions have common L, d, this is $C_e = 32L^2/(363 Q_e^2 d^3)$. We include one oriented row-zero event per row. The opposite column questions are retained for identification without a second row-zero test. This fixes the convention in the constant. Duplicating the four row-zero tests in the opposite orientation would multiply the constant by two.

Since the density factor I_q/q has trace one, no factor of q enters the loss. Thus $\omega_q(G_e) = 1$ for every e, proving the promise in Theorem A. This calculation uses four rows, rather than suppressing them inside an unspecified continuity estimate.

9 Exact Bell attainment

Let $L(p) = \sum_{xyab} c_{xyab} p(a, b \mid x, y)$ have rational coefficients and let $C_{\mathrm{abs}} = \sum |c_{xyab}|$. Its value on any strategy lies in $[-C_{\mathrm{abs}}, C_{\mathrm{abs}}]$. Finite Schmidt truncation shows that its supremum over spatial strategies equals its supremum over finite-dimensional strategies.

9.1 A specified global bound

Assume first that $\beta \in \mathbb{Q}$ is a global upper bound. For fixed measurements the Bell operator

$$B(E, F) = \sum c_{xyab} E_{a|x} \otimes F_{b|y}$$

satisfies $B \le \beta I$. A normal state attains β only if every positive-weight pure component does so. For a pure vector this is equivalent to $B\psi = \beta\psi$, because $\beta I - B$ is positive. Compress to its Schmidt supports and normalize the largest amplitude as before. Vectorization yields the homogeneous equation

$$\sum c_{xyab} E_{a|x} D_{\mathrm{Sch}} F_{b|y}^{\mathsf{T}} = \beta D_{\mathrm{Sch}}. \tag{47}$$

At stage n use the variables of (3) and require every squared entry of the residual in (47) to be at most $C_{\mathrm{abs}}^2 \kappa/(n+1)$. Compression of a true witness satisfies this bound: only the

compact tails of the sandwiched terms contribute, with coefficient sum C_{abs}. Conversely the compact-sandwich lemma gives an exact nonzero limit solution. Its vectorization is an attaining strategy. Thus attainment is $\exists\kappa\,\forall n$ of a decidable predicate.

If validity is not promised, impose also $\omega_q(L) \le \beta$. This is Π_1^0: for each finite dimension, the universal assertion that all states and POVMs give value at most β is decidable by real quantifier elimination. Finite-dimensional density states suffice for this validity test. The conjunction remains Σ_2^0.

9.2 No bound supplied

The quantifier pattern is explicitly $\exists\kappa \ge 1\,\forall n \ge 1\,\mathcal{R}_{\mathrm{Bell}}(L,\kappa,n)$, with $\mathcal{R}_{\mathrm{Bell}}$ decidable as follows. At stage n introduce also a real variable $\beta \in [-C_{\mathrm{abs}}, C_{\mathrm{abs}}]$, use the same residual bound, and require $\beta \ge \omega_n(L)$, where ω_n is the maximum in local dimension n. This latter requirement is a universal semialgebraic formula in β, not a numerical oracle. Full real quantifier elimination decides the stage formula with its existential and universal real variables.

An attaining strategy supplies all stages. Conversely pass to a subsequence with $\beta_n \to \beta_*$, $D_{\mathrm{Sch},n} \to D_{\mathrm{Sch}}$ in norm and effects weakly convergent. For every fixed dimension d, all sufficiently late stages have $\beta_n \ge \omega_d(L)$. Hence $\beta_* \ge \sup_d \omega_d(L)$. The limit equation (47) gives a unit vector attaining β_*. Its expectation cannot exceed that supremum, so equality holds. This proves Σ_2^0 membership for attainment of the unspecified supremum.

9.3 A specified scalar value

Let $t \in \mathbb{Q}$ be supplied without a bound promise. A value strictly between the infimum and supremum has finite-dimensional values on either side, by equality of the finite and spatial extrema. A finite direct-sum mixture then attains t. Existence of a finite-dimensional strategy with value exactly t is Σ_1^0, by dimension search and quantifier elimination. Every remaining attainable value is an endpoint. Upper endpoints are handled above; lower endpoints are upper endpoints for $-L$. This finite union is Σ_2^0.

For hardness in all three cases use the winning-probability functional of G_e. Its coefficients are rational, its global bound and finite-dimensional supremum are both one, and attaining the specified value one is precisely perfect spatial play. Theorem 7.1 and Section 8 therefore prove Theorem B. In its promised-bound version every reduction output has valid bound 1, and the upper predicate restricts to the valid-bound domain, so the same map establishes promise-problem completeness.

10 Discussion and open problems

The hierarchy separates four ways to ask for certainty. Finite-dimensional perfection has a finite existential witness. Commuting perfection is controlled by compactness. Value one asks for arbitrarily accurate finite witnesses. Spatial perfection combines an unbounded finite choice with an infinite verification. Here the chosen object is the order of a clock phase.

The entropy restriction does not lower the complexity. Each YES instance G_e of our reduction has a finite-entropy witness. There is no computable entropy budget sufficient for every finite-entropy YES game: such a budget $B(G)$ would make $\mathrm{PERF}_{\mathrm{finS}}$ a Π_1^0 set by the uniform budget predicate in Section 3, contradicting its Σ_1^0-hardness from perfect completeness of $\mathsf{MIP}^* = \mathsf{RE}$. The analogous impossibility for a sufficient largest-Schmidt stratum is Theorem C. The fixed-κ compactness result does not give a uniform modulus over games. It also concerns zero constraints and extremal scalar eigen-equations, not a full prescribed correlation table.

10.1 Prescribed tables and escaping mass

For a concrete obstruction to the latter extension, put

$$D_{\mathrm{esc},n} = \mathrm{diag}(1, \underbrace{n^{-1/2}, \ldots, n^{-1/2}}_{n \text{ entries}}, 0, \ldots).$$

Then $\operatorname{Tr} D_{\mathrm{esc},n}^2 = 2$ and $D_{\mathrm{esc},n} \to |0\rangle\langle 0|$ in operator norm. Alice measures $|0\rangle\langle 0|$ and its complement; Bob has one trivial outcome. Every normalized vectorization of $D_{\mathrm{esc},n}$ gives Alice probabilities $(1/2, 1/2)$. The normalized limit gives $(1, 0)$. The largest-amplitude normalization preserves zero equations while changing nonzero normalized probabilities.

Unproved remarks on prescribed-table bounds. The following upper-bound remarks are not proved in this paper and are not used in any theorem. For a prescribed table with rational or real-algebraic entries, direct coding of countable operators and a normal vector gives a Σ_1^1 upper bound. A supplied computable Schmidt-tail modulus makes the normalized state family effectively compact and gives a Π_1^0 upper bound. An existential choice from a uniformly total computable family of such moduli gives Σ_2^0. Finite entropy also gives Σ_2^0, by (5) with an existential integer budget and rational slack. If an arbitrary program for a computable modulus is existentially supplied, totality and the tail conditions give a Σ_3^0 upper bound. These proposed bounds concern the respective restricted realization classes; establishing their effective coding details is outside the present proofs. We do not prove that every algebraic spatial table has a computable-tail realization, or give a completeness classification for unrestricted table membership.

10.2 The Fu–Miller–Slofstra remark

After Theorem 1.1, Fu, Miller and Slofstra state that their fixed-scenario membership problem is in RE for $t = q$ or $t = qs$ [10]. We found no proof of the general qs upper assertion in that discussion. The following consequence needs a uniform interpretation over finite scenarios.

Suppose membership of real-algebraic tables in C_{qs} has a uniform RE procedure. If every spatially perfect game in our reduction had an algebraic perfect table, enumerate all algebraic tables of its scenario, test the losing zeros exactly, and dovetail the membership procedures. This would enumerate precisely the YES inputs of a Σ_2^0-complete family, a contradiction. Under that interpretation, either the RE remark fails for qs, or some spatially perfect games in this family have no perfect correlation table with all entries algebraic. The conclusion is existential over YES games; it does not say that every YES game has only nonalgebraic perfect tables.

The literal remark is per fixed scenario. The scenario of G_e grows with e, which is hardwired into the CA alphabet, so separate nonuniform RE procedures for each scenario give no contradiction. Tension arises only if there is one procedure uniform across scenarios, together with algebraic perfect tables for every YES game of the reduction. This is an open discrepancy, not a refutation. Resolving the uniformity and algebraicity questions is a natural next problem.

10.3 The von Neumann type ladder

Version 1.0.0 of this paper asked how perfect play behaves when the ambient local algebras have type II or type III. Three papers by the authors answer them. The results below are proved there, not here. In each, a strategy is perfect normal play on an ambient von Neumann algebra of the stated type, with the commuting partner specified in the cited paper. The spatial strategies of Theorem A are the type-I case.

A single finite game already separates type III from semifinite hosts. The game G_{III} of [6, Theorem 1.1] has 45 Alice questions, 46 Bob questions and 243 uniformly sampled question pairs. It has a perfect normal strategy on the injective type $\mathrm{III}_{1/2}$ factor, and no perfect strategy has a normal marginal on a semifinite ambient algebra of either party. In particular it has perfect commuting play but no perfect spatial strategy. For factors with separable predual, hosting G_{III} is equivalent to containing an expected copy of the injective type III_1 factor. Injective hosts are excluded by a different game. The game of [5, Theorem 1.1] has a perfect correlation in the closure of the finite-dimensional correlations, realized on a matrix amplification of a type $\mathrm{III}_{1/2}$ factor. No perfect strategy admits a normal injective host for either player. Normal hyperfinite strategies therefore approach perfect play for this game but never attain it.

The arithmetical classification is [7, Theorem A]. For finite rational games, perfect normal play in a type II_1 algebra, in a type II_∞ algebra, or in some semifinite algebra is Σ_2^0-complete; the upper bound is Theorem 2.2 of [7], and the hardness reduction is Theorem C there. For each fixed $\lambda \in [0, 1]$, the type III_λ language equals commuting perfection [7, Theorem 7.2] and is therefore Π_1^0-complete. Commuting perfection without semifinite perfection is Π_2^0-complete, with hardness on a computable family of commuting-perfect games of quantum value one. The type-I language, whose classification is Theorem A of the present paper, and the type II_1 language are incomparable, and both are proper subsets of the type II_∞ language. Theorem 9.1 and Corollary 9.2 of [7] show that two directional differences are $D_2(\Sigma_2^0)$-complete: perfect play in type I but not in type II_1, and perfect spatial play attainable only with infinite entropy. Completeness of semifinite-but-not-type-I perfection, and of its finite-entropy variant, remains open there; both are Σ_2^0-hard and Π_2^0-hard. These are existence languages, in which the ambient algebra and its normal state are chosen together, with the conventions fixed in [7, Section 1]. Separations involving spatial play there use the reduction of Theorem 7.1.

10.4 Other questions

Can the game compiler be made substantially smaller? Can the period reader be realized by a simpler predicate? What are the optimal effective bounds when one supplies an entropy budget or a Schmidt-tail modulus? The finite-entropy and arbitrary-spatial decision sets have the same completeness degree, although the sets themselves differ: the entropy-game construction in [8] has spatial perfection and no finite-entropy perfect strategy. Their difference, the set of games with perfect spatial play but no perfect spatial strategy of finite entropy, is $D_2(\Sigma_2^0)$-complete [7, Corollary 9.2]; in particular it belongs to neither Σ_2^0 nor Π_2^0. A classification of prescribed-table membership would require additional control of the normalization that escapes in the example above.

How this work was done

The authors selected the questions and proof standards, directed the research, and take responsibility for every claim. AI systems from more than one developer assisted with exploration, derivation, programming and drafting.

The mathematical evidence consists of the proofs in this paper and the explicitly cited external theorems. The ancillary programs provide reproducible finite checks. Before the rewrite, review by a different model family checked the underlying derivations of the upper bound, compiler, rooted computation, clock, reduction, value-one argument and Bell attainment. The final manuscript was audited separately; the present revision addresses that manuscript audit. The finite programs test truncated tree and clock relations, a pulse-reading front, good primitive phases, and the compiler ingredients described in Appendix A. Their arithmetic and

scope are stated there; numerical checks are distinguished from exact checks.

These finite checks do not prove the infinite-dimensional compactness argument, the general reduction, or an unrestricted robustness theorem by sampling. The asymptotic conclusions rest on the written proofs. This preprint has not been peer reviewed.

A Reproducibility

The accompanying `verification/` directory contains the tree-and-clock checker, the primitive-phase checker, their execution logs, and the independent compiler replication artifacts available at packaging time. Its README states commands, dependencies, provenance and limitations. `SHA256SUMS` identifies every bundled file apart from itself. Public paths are relative. The three compiler entry points accept `-output-dir DIR`, defaulting to `compiler/generated/`; they refuse to overwrite the shipped compiler directory or toolkit. Windows memory telemetry is guarded on other platforms. Fresh revision logs and the check summary are in `logs/revision-checks/`; generated output directories and Python bytecode caches are excluded from the shipped-file manifest.

The numerical tree checker constructs the truncated Fock model, Weyl pair, coordinate increments, clock, counter and exact-return candidate. Its pulse-driven cellular automaton is a small front-and-mark model. It is not an implementation of every Turing-machine decoder used by the reduction. The checker tests the creation and range identities, modular scaling, endpoint recurrences, clock and reader equations, tableau propagation, centralizer commutators and top-depth defects. The four diagonal isometry equations fail exactly at the top depth, as predicted by (43); all other reported residuals are at floating-point precision. The analytical mass is (42).

The primitive-phase checker tests the existence of a good primitive phase for every prime below 4000 and tests the chord lower bound only for primes below 400. Its trigonometric evaluations use floating point. Lemma 6.1 is the proof for all primes. The independent exact tree suite checks ten models, depths 0 through 4 and clock orders $2, 3$, with 2000 exact checks in total. A separate minimal compiler instance has four bridges and one modular isometry, but no transported endpoint labels. It has 748 rational rows and 27 991 questions per party. Its independent structural audit checks 77 750 serialized bindings, 11 652 column-graph rules, all 748 row expansions, 333 scalar pair contexts and 2472 old forbidden events. It does not instantiate the full ten-label computation compiler. This minimal instance is the only compiled game: the full reduction game G_e of Theorem 7.1 was not compiled. These exact replication calculations and their relative-path packaging are separately identified in the ancillary README. No proof-assistant certification is claimed.

The paper is built with `python build.py` from `paper/`. This runs `pdflatex`, BibTeX and the required repeat passes, and fails if the final log contains a warning, an undefined reference, or an overfull or underfull box. The plain-text abstract is `abstract.txt`.

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